

Zurich ServiceNow AI Platform Capabilities

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CMDB data management

The integrity and health of the CMDB is essential for the various features that depend on the data in the CMDB. As the CMDB grows and infrastructures change, the CMDB can accumulate stale, duplicate, or outdated CIs and therefore no longer accurately reflect the IT infrastructure and applications in the organization.

The CMDB Data Manager is an essential tool where you can create, publish, and manage policies that reflect organizational needs for data management. The CMDB Data Manager is a comprehensive and integrated solution that scales to large CMDBs. It supports bulk management of CI life cycle operations such as deletion, archival, and attestation. Use the CMDB Data Manager to automate and govern CI life cycle operations to help maintain the CMDB in a healthy and efficient operational state.

Managing duplicate CIs is also an important data management task that helps maintain the health of the CMDB.

Working with CMDB Data Manager

CMDB Data Manager is a policy-driven framework for bulk management of CI life-cycle operations such as deletion, archival, and attestation. The CMDB Data Manager is a comprehensive and integrated solution that scales to large CMDBs and copes with rapid changes in a cloud-based world.

Large CMDBs can over time accumulate large amounts of stale CIs that can impact overall performance. Custom mitigating solutions can be difficult to develop and to maintain, and are also prone to errors. The CMDB Data Manager is the tool where you can create, publish, and manage policies. Create policies to automate and govern CI life-cycle operations to help maintain the CMDB in a healthy and efficient operational state.

Use the CMDB Data Manager to create policies that represent your organizational processes for managing the life cycle of CIs such as:

- Retire all computers without owners that were created more than a year ago ('Retire' policy type).

- Archive all Linux servers in the Seattle datacenter that haven't been updated for six months ('Archive' policy type).
- Delete all containers that haven't been discovered in the past week ('Delete' policy type).
- Attest all the CIs in a specified location (Attestation policy type).
- Approve cascade-delete, archive, or retire life-cycle tasks generated by [dependent CI management](#).
- Delete orphan, stale, or irrelevant records in non-CMDB related tables. The non-CMDB Related tables in the Related Entry [cmdb_related_entry] table have references to CMDB tables. A CI in a related table can, for example, become orphan if the referenced CI in the CMDB is deleted ('Delete CMDB Related Entry' policy type).
- Certify that attributes have a specific value required for the business.

You can apply CMDB Data Manager policies only to classes that extend the Configuration Item [cmdb_ci] table.

CMDB Data Manager experience in CMDB Workspace

You can use the [CMDB Workspace](#) landing page and its views to fully administer CMDB Data Manager, access high-level analytic and counts for its policies and tasks, and review your tasks. For information about using the CMDB Data Manager in CMDB Workspace, see [CMDB Data Manager experience in CMDB Workspace](#).

Terms

Policy

A CMDB Data Manager policy captures the overall management plan for a life-cycle event, such as CI retirement. A policy is associated with a subflow (the policy subflow) which creates the tasks (the policy tasks) for the target CIs of the policy. A policy is configured with a policy type and the policy tasks perform operations associated with that policy type, such as archiving or deleting a CI record. Also, you can configure a policy to require an approval.

The policy type, policy subflow, and policy tasks are all aligned to a specific CI life-cycle event. For example, a policy set with the delete

policy type is associated with the delete subflow, and its policy tasks handle the deletion of CIs.

A daily scheduled job processes all published CMDB Data Manager policies.

Policy subflow

The policy [subflow](#) contains the underlying logic to process a life-cycle event such as retire or delete. If the policy is configured to require approval, then the policy subflow runs only after a policy task is approved.

The base system provides several common subflows such as delete, archive, and retire, which you can use with policies. You can also create custom subflows for your organization.

Policy task

A separate task is created and assigned to each unique Managed By Group value within the set of target CIs in a policy. A policy task triggers the policy subflow, tracks the set of target CIs for the task, and handles the approval of the task, if necessary.

If a policy requires an approval, the policy tasks don't trigger the policy subflow until a member of the group assignment in the Managed by Group attribute of the target CIs, approves the tasks. If a task is rejected or if the Managed by Group attribute is empty, the task is assigned to an administrator who must manually intervene to resolve the task.

If a policy isn't configured to require an approval, then the policy tasks are automatically approved.

CI exclusion list

A set of CIs to which policies of a specified type don't apply.

Policy types

You can create policies of the following types:

Delete

Use to remove a CI from its current table with no option to restore the CI into an active state.

Retire

Use to retire a CI while keeping the CI active in list views and in processes such as CMDB Health.

Attestation

Use to assign and process attestation tasks that verify the existence of actual IT infrastructure and applications that you own. As CIs are continuously ingested into the CMDB from various data sources, attesting CIs support the integrity of the CMDB. For more information about using the Attestation policy type, see [CIs attestation](#).

Archive

Use to remove a CI from its current table and store the CI in a separate archive table for temporary retention. Archiving a CI excludes the CI from views and from features such as maps and the relations formatter. During the retention period, you can [restore CIs into active state](#). At the end of the retention period, archived CIs are deleted from their archive table.

Delete CMDB Related Entry

Use to clean up any irrelevant or stale data from related tables to help keep CMDB data healthy and relevant as the state of referenced CIs change.

Related tables, such as the Serial Number [cmdb_serial_number] table, aren't part of the CMDB hierarchy but still qualify as CMDB data. Related tables don't inherit from the Configuration Item [cmdb_ci] table, but have at least one column that references a CMDB CI. Related tables are specified in the Related Entries [cmdb_related_entry] table.

Certification

Use to certify that specific attributes are of a specific value.

You can implement your Retire, Delete, and Archive policies so that they follow [Common Service Data Model \(CSDM\)](#) standards where for example, CIs are archived and deleted only when a CI is already in retired state. When you create these life-cycle policies, the system applies processes to manage any dependent CIs that might be left behind. For more details about these processes and about verifying that the feature is enabled, see [Dependent CIs management](#).

ServiceNow AI Platform® data archiving

The functionality that the Archive policy type in CMDB Data Manager provides, relies and extends the ServiceNow AI Platform® [data archiving feature](#), applied specifically for CMDB CIs. While processing an Archive policy to archive CMDB CIs, CMDB Data Manager uses components and processes of ServiceNow AI Platform® data archiving in the following ways:

- The Archive Rule [sys_archive] table contains the ServiceNow AI Platform® archive rules including the Archive Configuration Items CMDB archive rule which CMDB Data Manager Archive policies use.
- Data Manager relies on the Archive scheduled job to run (every hour by default) and process CMDB Data Manager archive policies. The Archive scheduled job is stored in the Schedule Item [sys_trigger] table.
- In the ServiceNow AI Platform® table Archive Job Execution Chunks [sys_archive_run_chunk], the Keys attribute contains the sys_ids of the CMDB CIs to be archived (where Rule ID is the CMDB archive rule ID).

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Archived records are stored in the ServiceNow AI Platform® archive tables, which are prefixed by 'ar_'. In a similar way, the first time that a CMDB archive job runs, it creates an archive table for each CMDB class (prefixed by 'ar_cmdb'). The initial CMDB archive task, therefore, takes longer than subsequent CMDB archive tasks.

For each Data Manager archive policy, the system batches the policy CIs to be archived into batches of 1000 CIs. The sys_archive_run_chunk table contains a record for each of the batches.

CMDB archive tables, such as ar_cmdb_ci_computer, are listed under **All > System Archiving > Archive Tables**.

When using the CMDB Data Manager to archive CIs, you can also directly apply ServiceNow AI Platform® data archiving features, such as [restore CIs into active state](#) during a CIs retention period.

CI life cycle management

You can use life-cycle policies in CMDB Data Manager to manage the life cycle of CIs. Those life-cycle policies use retirement definitions that specify the retirement state for classes in your organization and support

the transition of CIs through life cycle stages. For more information about accessing and managing retirement definitions, see [Retirement definitions](#).

In general, CIs that are no longer needed should be retired. Then, to complete the CI life-cycle, retired CIs should be deleted or archived according to business needs. The following high level steps describe how to manage CIs' life cycle:

1. Use a non-production instance as a safe environment for configuring and testing life-cycle management in your organization.
2. Choose the CI class for which you want to define a retirement definition while carefully considering derivation. Due to derivation, the retirement definition that you are specifying for a class, is also in effect for all child classes that don't have their own retirement definition.
3. Specify the retirement definition for the class.
4. Create a retirement policy targeting the CIs that you want to retire.
5. Create a delete or an archive policy targeting the retired CIs.
6. After testing and verification that the entire life-cycle management plan works as intended, transfer all of those configurations to the production instance.

The life cycle state of a CI affects the CIs visibility and inclusion in ongoing CMDB processes:

- A retired CI isn't excluded from any views or processes such as CMDB Health.
- An archived CI no longer exists in its active table and instead it's stored in a separate archive table. Archived CIs are no longer visible or included in processes such as list views, maps, and relations formatters. Archived CIs can be retained for a specified retention period before being deleted from the archive table. During the retention period, archived CIs can be manually restored into an active state by using the ServiceNow AI Platform® feature to [restore CIs into active state](#).

Note: Deleting a CI is an irreversible operation. A deleted CI no longer exists in the table that it belonged to and there's no way to restore it to an active state.

Configure the environment for CMDB Data Manager

Prepare your environment for using the CMDB Data Manager:

1. Some policy types such as the life-cycle policies Retire, Archive, and Delete, require that an active [retirement definition](#) exists for each targeted class in the policy. This requirement doesn't apply to all policy types. For example, this requirement doesn't apply to the Attestation policy type. If you attempt to create a policy of a policy type for which this requirement applies but isn't met, an error message appears and the operation fails.
2. You can streamline approval of policies by populating the Manage by Group attribute of CIs that you plan to target in policies. Use the CI Class Manager to populate that attribute for an entire class, in a single synchronization operation. For more information about this data synchronization, see [Set the group for a CI or an entire class of CIs](#). If the Managed by Group attribute isn't populated for a CI, then the approval process is directed to the administrator.

CMDB Data Manager experience in CMDB Workspace

Use the CMDB Data Manager in CMDB Workspace to administer policies, CIs life-cycle and attestation tasks, view analytics about on-going CMDB Data Manager processes, and to approve or reject assigned tasks.

Use the [CMDB Workspace](#) landing page and its views to administer CMDB Data Manager, to access overall analytic and counts for its policies and tasks, and review your tasks:

- Use the [Management view in CMDB Workspace](#) to create and track policies, and to manage exclusion lists and retirement definitions.
- Use the Important actions tile on the CMDB Workspace landing page to access data attestation and life cycle approval tasks such as reassignment requests and unassigned overdue tasks. Drill down these task cards to see further details about the tasks.
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Use the My work tile on the CMDB Workspace landing page to access all open tasks, which are assigned to you, or to the group assigned in the Managed by Group attribute and which you're a member of. For CMDB administrators, any unassigned tasks are also included for reassignment. Review and process all tasks. Review and process attestation tasks by checking the physical existence of IT infrastructure or applications associated with the CIs in the tasks.

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Use the [My Work view in CMDB Workspace](#) to access attestation and non-attestation tasks that are assigned to you or to an assignment group that you belong to in accordance with CMDB Data Manager policies. Overdue attestation tasks appear in a separated **Overdue tasks** list.

Manage policies in CMDB Data Manager in CMDB Workspace:

- [Create a CMDB Data Manager policy](#)
- [Review CMDB Data Manager tasks](#)
- [Review CMDB Data Manager attestation tasks in CMDB Workspace](#)
- [Manage retirement definitions for CMDB Data Manager](#)
- [Manage exclusion lists for CMDB Data Manager](#)

Administer CMDB Data Manager

As a CMDB administrator, you can create policies, manage CI attestation, manage CI exclusion lists, access CMDB Data Manager insights, and access subflows and settings that are used in the CMDB Data Manager in CMDB Workspace.

Get started with CMDB Data Manager

The Data Manager overview page in CMDB Workspace provides a dashboard view of Data Manager tasks, and lets you get started in using the CMDB Data Manager.

Before you begin

Role required: sn_cmdb_admin

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
2. Select the **Data Manager** link in Management tools, in the Manage section.
3. Select **Data Manager overview** in the left navigation bar.
The Data Manager overview page provides the following cards:

Open tasks

Count of the open tasks created for Data Manager policies.

Drill down the card to open a list view with details of the tasks associated with the card.

Unassigned tasks

Count of the open tasks created for Data Manager policies, which aren't assigned.

Drill down the card to open a list view with details of the tasks associated with the card.

Pending tasks

Tasks in Pending state, such as archive tasks where archive is still being processed.

Closed incomplete tasks

Tasks that encountered errors while processing, resulting with the task not completing successfully. Such tasks are in a Closed Incomplete state.

Overdue tasks

Count of tasks with a due date, such as attestation tasks, whose due date has passed.

Aging tasks

A bar chart showing counts of tasks without a due date and with no activity for over the number of days specified by the `sn_cmdb_ws.glide.cmdb.data.manager.aging.policy.task.timeframe` system property (30 by default).

Policies failing evaluation

Policies whose evaluation failed at some point and are in an error state. These policies require your attention, and after resolving any issues, you can publish them again.

Workload

- Groups: Tasks that are assigned to a group.
- Individuals: Tasks that are assigned to individual users.

What to do next

- [Create a CMDB Data Manager policy.](#)
- [Manage retirement definitions for CMDB Data Manager.](#)
- [Manage exclusion lists for CMDB Data Manager.](#)
- [View CMDB Data Manager analytics.](#)
- View properties and other settings for CMDB Data Manager by selecting **Settings** in the left navigation pane. For details about the system properties associated with those settings, see [Components related to CMDB Data Manager](#).

Manage retirement definitions for CMDB Data Manager

Use CMDB Workspace to create, delete, edit, or activate retirement definitions in your organization, for a class. Tables that are targeted in CMDB Data Manager life-cycle policies, must be associated with an active retirement definition. Target CIs in Retire policies, are set to match their associated class retirement definitions.

Before you begin

Role required: sn_cmdb_admin

About this task

Some retirement definitions in the base system are inactive and you must activate any retirement definition that you want to use with a CMDB Data Manager life-cycle policy by setting its Active attribute to **true**. Retirement definitions are stored in the CMDB Retirement Custom Definitions [cmdb_retirement_custom_definitions] table.

When a CMDB Data manager retires a CI, ensure that any of its dependent CIs are also retired and don't remain as orphan CIs, by enabling management of dependent CIs. Orphan CIs can impact the consistency, accuracy, and integrity of the CMDB.

- For information about enabling and managing dependent CIs, especially in the context of cascade-retire, cascade-archive, and cascade-delete, see [Dependent CIs management](#).
- For details and definitions of independent and dependent CIs, see [CMDB classifications and class dependency](#).

Note:

- Always be aware of the automatic derivation of retirement definitions throughout the class hierarchy. For example, an active retirement definition for the Configuration Item [cmdb_ci] class can potentially be in effect throughout the entire CMDB hierarchy (aside from classes with their own retirement definition).
- You can't deactivate (or delete) the retirement definition of the Configuration Item [cmdb_ci] class.
- Each CMDB class can be associated with only a single retirement definition.

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select the **Data Manager** quick link on the Home view.

2. In the Get started area on the Data Manager overview page, select **Manage retirement definitions**.
3. On the Data Manager settings page, review the Retirement definitions list view.
 - a. (Optional) Drill down to a definition and then, on the Retirement definition form, toggle the **Active** switch to activate or deactivate a definition.
 - b. (Optional) Edit the conditions on the definition.
 - c. Select **Save definition**.
4. On the Data Manager settings page, select **Add definition**.
5. Fill out the Retirement definition form and then select **Save definition**.

Field	Description
Active	Toggles the retirement definition between active and inactive.
Table	Table that the retirement definition applies to. The retirement definition will be derived by all child classes of the specified parent class.
Condition builder	Condition that reflects on what retirement means in the organization. Target CIs in Retire policies that belong to the specified table are set as specified in the retirement definition. You can, therefore, only use AND clauses in the definition. OR clauses aren't applicable.

Delete a CMDB Data Manager retirement definition

Delete a retirement definition that is no longer needed by CMDB Data Manager, in CMDB Workspace.

Before you begin

The following restrictions are in effect:

- The retirement definition that you want to delete must be in an inactive mode (Active = **false**).
- The retirement definition for the Configuration Item [cmdb_ci] class can't be deleted.

Role required: sn_cmdb_admin, data_manager_admin

About this task

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select the **Data Manager** quick link on the Home view.
2. In the Get started area on the Data Manager overview page, select **Manage retirement definitions**.
3. On the Data Manager settings page, in the Retirement definitions tile select the retirement definition that you want to delete.
4. On the Retirement definition page, select **Delete definition**.
5. In the Deleted retirement definition dialog box, select **Delete**.

Result

After the deletion, a retirement definition for the class is derived from a parent class.

Manage exclusion lists for CMDB Data Manager

Create and manage exclusion lists for the various policy types used in CMDB Data Manager, in CMDB Workspace. Policies of the specified type won't target CIs and other records in the exclusion list for that policy type.

Before you begin

Role required: sn_cmdb_admin

About this task

Note: The Certification policy type doesn't support exclusion lists.

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select Management in the CMDB Workspace menu bar.
2. Select the **Data Manager** link in Management tools, in the Manage section.
3. Select **Excluded records** in the left-side bar.
The Excluded records list view shows lists of all records currently excluded, grouped by policy type. You can expand any group to show its CIs and other records and then drill down to any record.
4. (Optional) Remove excluded records from the exclusion list:
 - a. Expand a policy type group to show its list of excluded records.
 - b. In the list view, select the records that you want to remove from the exclusion list.
 - c. Select **Remove from list**
 - d. In the Confirm removal from exclusion list dialog box, select **Confirm**.
The selected records are removed from the exclusion list and can be targeted for the policy type.
5. (Optional) Add records to an exclusion list:

- a. Select **Exclude Records** and fill out the Data Filter form.

Field	Description
Tables	Table from which to select CIs or records for the exclusion list.
Filter conditions	Use the condition builder to specify the criteria that CIs or records from the specified Tables must meet to be included in the exclusion list.
Related List Condition	Add a condition that is based on related lists that are associated with the records that you want to exclude.

- b. Select **Apply filters** and then review the Search results list.
- c. Select the records that you want to add to exclusion lists and then select **Continue**.
- d. On the Select policy type page, select the policy types for which to exclude the selected records and then select **Continue**.
- e. Review the details of the exclusion on the summary page and then select **Save**.

Create a CMDB Data Manager policy

Create a CMDB Data Manager policy in CMDB Workspace to automatically process CIs life-cycle events such as deletion. Applying consistent and standard life-cycle policies to CIs helps maintain the health of the CMDB.

Before you begin

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The life-cycle policies Retire, Archive, and Delete, require that an active retirement definition exists for each targeted class in the policy. If you

attempt to create a policy of a policy type for which this requirement applies but isn't met, an error message appears and the operation fails.

For more information about retirement definitions, see [Retirement definitions](#).

- Verify that any custom subflow that you want to associate with a policy exists:
 - 1: Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
 - 2: Select the **Data Manager** link in Management tools, in the Manage section.
 - 3: Select **Subflows** in the navigation bar on the left.
 - 4: Review the subflows on the Data Manager subflows page, and drill down to a subflow to open it in Workflow Studio.
 - 5: Select **Create new subflow** to [create a new subflow in Workflow Studio](#).
- To require a review and an approval for a policy task: Ensure that the Managed By Group attribute is populated in target CIs and that the assigned users have the privilege to approve the policy tasks.
- When [Asset Management](#) is activated, check if there's an asset record associated with that CI before retiring the CI. Check the associated asset record, if there's one, and verify that its asset state (install_status) is Retired.

Role required: sn_cmdb_admin

About this task

Specify for each policy a policy type, a subflow for life-cycle policies, a set of CIs to operate on as target CIs, and any other options specific to the selected policy type. Target CIs must typically be from classes extending the Configuration Item [cmdb_ci] table, and with the Delete CMDB Related Entry policy type, also tables in the Related Entries [cmdb_related_entry] table. However, the Certification policy type for example, also supports non-CMDB tables. Many fields dynamically appear on the form based on how you set previous fields and some fields are specific to the policy type that you set.

Set condition filters to specify the initial set of CIs that the policy applies to. You can then further narrow down the initial set of CIs by using a CI exclusion list for the policy type. During the final preview of the policy, or from a policy task, you can select individual CIs to also exclude for the policy type. In addition, for operations that require that CIs are in a retired state, retirement definitions are enforced. The policy eventually applies to the resulting set of CIs, after applying all those filters.

Note: CMDB Data Manager limits the number of target CIs per task to 10,000. Therefore, when a task exceeds that number, Data Manager automatically creates as many additional tasks as needed to include all the CIs for the task. For example, if you target 30,000 CIs in an attestation task, Data Manager breaks down that task into three tasks, each targeting 10,000 CIs.

For more information about life-cycle state definitions and other concepts for working with the CMDB Data Manager, see [Working with CMDB Data Manager](#).

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select Management in the CMDB Workspace menu bar.
2. Select the **Data Manager** link in Management tools, in the Manage section.
3. Select **Policies** in the left-side bar.
For information about the Policies page, see [View CMDB Data Manager policies](#).
4. Select **Create new policy**.
5. Fill out the General information form and then select **Continue**.

Field	Description
Name	Unique name for the policy.
Description	Description for the policy.

Field	Description
Policy type	Life-cycle event or data management action, such as Delete or Attestation, that this policy manages, indicating the type of actions to perform on target CIs. Your Policy Type setting determines which fields appear in the forms that follow.
Work notes	Notes that are internal and visible only to administrators.

6. Fill out the Data Filter form.

- a. Fill out filter conditions for the target CIs and records for the policy.

Field	Description
Table	Table from which to select target CIs or records for the policy. Only classes that you have read access to, are available for selection.
Filter conditions	Use the condition builder to specify the criteria that CIs from the specified Table must meet to be included as target CIs for the policy. Select New condition set to add a condition to the filter.

Field	Description
Related List Condition	Add a condition that is based on related lists that are associated with the target records for the policy.

- b. Select **Apply filters** and then review the results of the impact analysis.
This analysis estimates the number of CIs that the policy applies to based on the policy filters, any CI exclusion lists, and the life cycle stage of CIs. For example:

- If the policy type is retired, CIs that meet the policy filters but are already in a retired state according to the class retirement definitions, aren't targeted for the policy.
- If the policy type is archive or delete, CIs that meet the policy filters but aren't retired according to the class retirement definitions, aren't targeted for the policy. A filter on the Life Cycle Stage or the Life Cycle Stage Status attributes has no effect if the respective class retirement definition is based on those attributes.

Review the lists in the following tabs:

- **Included records:** You can select records to be excluded from the policy, and select **Exclude records**.
- **Excluded records:** Select records to be included in the policy, and select **Include records**.

Note: The Excluded records tab doesn't appear when the policy type is Certification.

- c. Select **Continue**.

7. Fill out the Assignment form and then select **Continue**.

- a. Set the **Assignment type** and then the **Task assignment** field that appears, to specify the type of assignment and the table columns to use for assigning the policy tasks. The drop-down list in **Task assignment** is dynamically populated with columns in the table that the policy is created for, and that reference **Assignment type** tables, as follows:

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User Field: Assigns policy tasks to a user in the user field specified in **Task assignment**. The drop-down list in **Task assignment** contains fields in the table set by the data filter, that reference the User [sys_user] table (such as Attested by and Assigned to).

During assignment, records will be grouped based on the selected column, allocated to tasks, and then the tasks will be assigned to the resulting users.

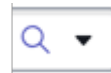
•

User Group Field: Assigns policy tasks to the user group in the user group field specified in **Task assignment**. The drop-down list in **Task assignment** contains fields from the table set in the data filter, that reference the Groups [sys_user_group] table (such as Change Group and Support Group).

During assignment, records will be grouped based on the selected column, allocated to tasks, and then the tasks will be assigned to the resulting user group.

- **Specific User:** Assigns policy tasks to the user specified in **Task assignment**. The drop-down list in **Task assignment** contains users with the data_manager_user role.
- **Specific User Group:** Assigns policy tasks to the user group specified in **Task assignment**. The drop-down list in **Task assignment** contains user groups that are either directly associated with the data_manager_user role, or contain at least one user with the data_manager_user role.

In the **User Group Field** and **User Field** you can use dot-walking by



selecting the search icon for the field: . In the selection dialog box, drill down lists by expanding or selecting list items that are links (usually appear in a unique text color).

- For certification and attestation tasks, set the **If task assignment field is empty** field to create assigned or unassigned tasks in cases where the specified task assignment field is empty. If you select to create an assigned task, then depending on the **Assignment type** setting, select the user or user group to assign a

task to in that situation. Unassigned tasks are later reviewed by an administrator for assignment.

8. Fill out the Options form and then select **Continue**.
The options that appear depends on whether they are relevant to the selected policy type. Therefore, some of the following options don't appear on your form. For example, not all policy types are associated with a subflow.

Field	Description
Days to complete	Maximum number of days that policy tasks must be completed by. If notifications are enabled for certification or attestation tasks, then this number is used to calculate the milestones for sending notifications. The full time (100%) interval starts when a task is created and ends when the number of days to complete the task arrives. Notifications are sent if a task isn't closed when 50%, 70%, and 90% of that interval passes. For more information about setting those notifications, see Components related to CMDB Data Manager.
Subflow	A subflow with the actions that will run on the target CIs for the policy. The subflow typically matches the policy type. For example, if Policy type is set to Delete CMDB Related Entry, then set

Field	Description
	Subflow to Delete Related Entry Configuration Item.
Needs review	Option to require a review and an approval of the policy tasks, by the group assignment in CIs' Managed by Group attribute or by an administrator. Otherwise, all policy tasks are approved automatically.
Retention time	The length of time for retaining archived CIs in the archive table before they're deleted. During the specified retention period, you can use the ServiceNow AI Platform® data archiving feature to restore archived CIs. Note: You can't use the CMDB Data Manager to re-archive a CI that was previously archived and restored.
Display fields	One or more fields that appear in list views in Data Certification tasks, that uniquely identify the records that require certification. The selected fields can't overlap fields in Certification fields.

Field	Description
Certification fields	One or more fields whose value requires verification and certification. The selected fields can't overlap fields in Display fields.
Allow field updates	Option to allow certification task reviewers to update field values while reviewing CIs. - When selected, certification task reviewers can update field values in order to certify a CI. - When clear, certification task reviewer can't update field values, and therefore reject CIs that aren't compliant.
Allow empty field values	Option to allow users to certify CIs with empty attributes: - When selected, certification task reviewers can certify or fail certification of a CI with an empty attribute. - When clear, certification task reviewers can fail certification of a CI with an empty attribute but aren't able to certify that CI.
Instructions	Any instructions to assigned users, to help them complete the tasks.

9. Fill out the Schedule form and then select **Continue**.

Field	Description
Schedule	Recurring schedule for how often and when to run the task. The On demand option doesn't set any recurring schedule for running the policy. To learn how to run a task configured with On demand, see 'What to do next' below. Your setting determines which other fields appear on the form.
Start time	Time in the day to start running the policy.
Day of the week	Day of the week to run the policy.
On	Day of the month to run the policy.
Start day and time	Date and time of first run of the policy.
Run after every	Interval (by days) between subsequent runs after running the policy for the first time.
Run policy scheduled job as	User to set the Run as field with, in scheduled jobs for attestation and certification tasks. By default, the drop-down list is populated with users in the Data Manager Scheduled Job

Field	Description
	Users user group and the field is set to the user specified by the Scheduled job default user property (DataManager Job Runner). For more information about this property and the Scheduled job default user user role, see Components related to CMDB Data Manager .

- Review the policy details on the Review form, and then select **Publish policy** to activate the policy or **Save & Exit** to save the policy as a draft.
You can later continue to configure a draft policy, and then publish it when it's ready.

Result

After you publish a policy:

- A daily scheduled job processes the published policy and policy tasks are assigned according to the policy settings. If the policy is associated with a subflow, then policy tasks trigger the policy subflow. Policy execution issues are recorded in an error log with notifications sent to the CMDB Data Manager administrator.

If the policy is configured to require an approval for its tasks, then email notifications are sent to members of the assignment group in the Managed by Group attribute of the CI. If the policy is associated with a subflow, then a policy task triggers the policy subflow only after the task is approved.
- If the policy is associated with a subflow, then after a policy task is complete, the policy subflow closes the task. For an Attestation policy (which isn't associated with a subflow), a user must process all CIs in the task and submit the task to close it.

- For Attestation policies, attestation tasks are assigned to users as specified, and those tasks appear in the [CMDB Workspace](#) when those users log in.
- For some policy types, such as **Delete**, the list of the target CIs is rolled up in a CSV file that is then attached to the task for tracking purposes.
- Stale tasks are set to **Closed Canceled** by a daily scheduled job. A task becomes stale when it's still open and not approved after at least 90 days. The number of days after which a task is considered stale is determined by the `cmdb.data.manager.stale.task.life.in.days` system property.

What to do next

- [Publish a draft CMDB Data Manager policy.](#)
- On the Policies page, in the Published policies list view, select a policy to see policy details on a policy form. Select **View Policy** to see the settings in all of the policy form pages.
- On the Policies page, in the Published policies list view, select a policy that you want to deactivate. On a policy form select **Deactivate Policy** to temporarily prevent the policy from running.
- Select **View Open Tasks** in the Open Policy Tasks tile to track the processing of policy tasks in the CMDB Data Management Task Control list view. The Success Percent column shows the percentage of CIs in the task for which the task is completed. A CI is counted as complete in an archival task only after the archival process has been fully completed for the CI (and isn't counted as complete while the CI is initially staged for archival for example).
- For **On demand** policies, the system generates a scheduled job prefixed with the policy name. To run such task:
 - 1: Navigate to **System Definition > Scheduled Jobs**.
 - 2: Select the scheduled job for the task policy that you want to run.
 - 3: On the scheduled job form, select **Execute Now**.
- Users log into [CMDB Workspace](#) to review and process attestation tasks assigned to them.

- Users log into [CMDB Workspace](#) to [review and process non-attestation tasks](#) assigned to them.

Publish a draft CMDB Data Manager policy

If you created a draft Data Manager policy, or imported legacy certification schedules, you can activate that draft policy by publishing it in CMDB Workspace.

Before you begin

Role required: sn_cmdb_admin

Procedure

1. Navigate to **All > Workspaces > CMDB Workspace** and then select Management in the CMDB Workspace menu bar.
2. Select the **Data Manager** link in Management tools, in the Manage section.
3. Select Policies in the left-side bar in the Data Manager overview page.
4. Select the Draft policies list view on the Data Manager policies page.
5. Select the policy that you want to publish and then, on the policy form, select **Edit Policy**.
6. Review any policy settings by selecting the pages to review in the left-side bar.
7. Select Review in the left-side bar and then, on the Review form select:
 - a. **Publish Policy** to activate the policy.
 - b. **Save & Exit** to save any changes and keep the policy in draft mode.

What to do next

Deactivate a policy to temporarily prevent the policy from running:

1. On the Policies page, select the Published policies list view.
2. Select a policy that you want to deactivate.
3. On a policy form select **Deactivate Policy**.

View CMDB Data Manager policies

View Data Manager policies on a dashboard view in CMDB Workspace, with a timeline of upcoming scheduled policies, counts, and grouped by policy status.

Before you begin

Role required: sn_cmdb_admin

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select Management in the CMDB Workspace menu bar.
2. Select the **Data Manager** link in Management tools, in the Manage section.
3. Select **Policies** in the left-side bar.
Select a tab on the Data Manager policies page:
 - Published policies: Provides the following cards:

Upcoming scheduled policies

Timeline showing scheduled policies across time. On the timeline, you can:

- Point to indicators to show details about a scheduled policy.
- Zoom in or out to extend or shorten the time span of the timeline.
- Go forward or backward in time.
- Show or hid the timeline legend, where you can toggle the Attestation Policy switch to show or hide attestation policy indicators on the timeline.

Policies

Pie chart showing the breakdown of policies by policy type.

Published policies

List view of policies that are published. You can drill down a policy to show the policy form view with further details about the policy.

- Draft policies: Policies in draft state.
- De-activated policies: Policies that have been de-activated and therefore aren't active.
- Policies failing evaluation: Policies that are invalid and therefore require your attention.

What to do next

- [Create a CMDB Data Manager policy](#)
- [Manage retirement definitions for CMDB Data Manager](#)

View CMDB Data Manager analytics

The CMDB Data Manager in CMDB Workspace provides charts and counts that show the overall state of Data Manager policies in the organization. Review these details to track progress and to identify any problems that require your attention.

Before you begin

Role required: data_manager_admin or sn_cmdb_admin

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select Management in the CMDB Workspace menu bar.
2. Select the **Data Manager** link in Management tools, in the Manage section.
3. Select **Analytics** in the left navigation bar.

The Data Manager analytics page provides the following charts:

CIs processed by lifecycle policies

Counts of CIs processed by each type of lifecycle policy type such as Delete and Archive, over the past six months.

Average time to close tasks

Average number of days it took to close tasks, by general policy type such as lifecycle and attestation, over the past six months.

Closed complete vs closed incomplete tasks

Number of tasks per month, that were closed complete and incomplete, over the past year.

Policy execution time

Number of policies per day, broken by the length of execution time, over the past month.

Review CMDB Data Manager tasks

Review Data Manager tasks (such as life-cycle tasks) that are assigned to you, to an assignment group that you belong to, to people you manage, or to members of a group that you are the manager of. All in accordance with CMDB Data Manager policies. During the review, you can approve, cancel, reject, or reassign a task.

Before you begin

Role required: `sn_cmdb_user`, `sn_cmdb_editor`, or `sn_cmdb_admin`

Canceling a life-cycle task is available only for tasks that are in the 'Not started' status, and the required role is `data_manager_admin` or `sn_cmdb_admin`.

About this task

- To review attestation tasks, see [Review CMDB Data Manager attestation tasks in CMDB Workspace](#).

- To review certification tasks, see [Review certification tasks in CMDB Workspace](#).

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **My Work** in the CMDB Workspace menu bar.
2. In the My tasks section, select the **All other tasks** tab.
3. Review the Approval tasks list view and then select the task that you want to process.
4. Review the task details on the task form.
5. (Optional) CMDB administrators can select **Cancel task** if the task isn't needed.
The canceled task is deleted without any further processing.
6. (Optional) Select **Reassign** if the task isn't appropriately assigned, and then select **Assign to me** or **Request reassignment** as appropriate.
7. (Optional) Select **Review records** to review the target CIs and records for the task.
 - a. Select any records that you want to exclude for the task and then select **Exclude**.
The records being excluded for the task are also being added to the exclusion list of the associated policy type. Therefore, these records will be excluded from any future tasks of the associated policy type.
 - b. Select **Reassign** if the task isn't appropriately assigned.
 - c. Select **Reject task** to reject a task, fill in a **Reason for rejection** and then select **Save**.

You can only reject a task that is assigned to you.

After you reject a task, the State of the task is set to **Closed Canceled** and Approval is set to **Rejected**. Also, appropriate notifications are sent to members of the group specified in the glide.cmdb.data_manager.rejected.task.notification_group system property. For more information about how this property

is used for notifications, see [Components related to CMDB Data Manager](#).

- d. Select **Assign to me** or **Request reassignment** as appropriate.
- e. Select **Approve task** to trigger the subflow associated with the policy.

Retirement definitions

Specify a retirement definition for a class, to reflect protocols in your organization applicable to the class and to its child classes. These definitions support the transition of CIs through life cycle stages as implemented by CMDB Data Manager life-cycle policies. A class can have a retirement definition by derivation, or by a retirement definition specified for that class.

CMDB Data Manager requirement and usage

An active retirement definition is required for each targeted class in a Retire, Archive, or Delete CMDB Data Manager policy. A retirement definition is used to flag a CI as retired according to the CI's class retirement definition in effect. When retiring a CI, the CI's attributes are set according to the retirement definition for the CI's class. A subsequent operation such as delete or archive, that needs to be applied to retired CIs, first checks if a CI is in a retired state by verifying that it satisfies the retirement definition for its class.

Retirement definitions are used in life-cycle policies as follows:

- Setting target CIs in Retire policies, according to their associated class retirement definitions.
- Filtering out target CIs in Retire policies, that are determined to already be in a retired state according to their associated retirement definitions. Those CIs aren't targeted even if they meet the policy filter.
- Identifying CIs as being ready to be processed by an Archive or Delete policies. Archive and Delete policies can only process CIs that are in a retired state that matches their associated class retirement definition. Therefore, target CIs in Archive and Delete policies, are checked against their associated class retirement definition to verify that they are in a retired state before they can be archived or deleted.

Derivation across the CMDB hierarchy

Retirement definitions are derived throughout the CMDB hierarchy in the same way that rules, such as identification rules, are derived. Child classes extended from a parent class with a retirement definition, derive that definition unless there's a retirement definition defined at the child class level.

When retirement definitions for classes exist in different levels in the class hierarchy, then the retirement definition that applies is the one that is most specific for a class. For example, if there are retirement definitions for both, the Computer and the Configuration Item class, then the Computer class definition also applies to its descendant classes. All other classes that aren't descendants of the Computer class, use the Configuration Item retirement definition.

Always be aware of the automatic derivation of retirement definitions throughout the class hierarchy. For example, an active retirement definition for the Configuration Item [cmdb_ci] class can potentially be in effect throughout the entire CMDB hierarchy (aside from classes with their own retirement definition).

Predefined retirement definitions

The base system includes predefined retirement definitions for key classes such as Hardware [cmdb_ci_hardware] and Application [cmdb_ci_appl], and are stored in the CMDB Retirement Custom Definitions [cmdb_retirement_custom_definitions] table. For example, the predefined retirement definition for the Service [cmdb_ci_service] class specifies that for a retired CI, the value of the attributes [operational status], [Phase], and [Status] is **Retired**.

Retirement definition for the Configuration item [cmdb_ci] class behaves in the following important ways:

- It is active by default and you can't deactivate it.
- You can't delete it.
- By default, it is in effect throughout the entire CMDB hierarchy due to derivation.

The rest of the predefined retirement definitions are inactive by default, and you must activate a retirement definition that corresponds to a targeted class in a Retire, Archive, or Delete CMDB Data Manager policy.

You can use the default retirement definitions, or customize a definition to reflect practices in your organization. You can also add a custom retirement definition for a class. However, each CMDB class can be associated with only a single retirement definition.

To manage or delete a retirement definition, see [Manage retirement definitions in CMDB Workspace](#).

Duplicate CIs remediation

Duplicate CIs in the CMDB unnecessarily overload the system and interfere with the CMDB's integrity and reliability. It is therefore important to regularly use CMDB tools that detect and remediate duplicate CIs.

De-duplication tasks

When Identification and Reconciliation Engine (IRE) processes detect duplicate CIs, IRE groups each set of duplicate CIs into a de-duplication task for review and remediation. De-duplication tasks provide details about the duplication, including a list of all the duplicate CIs in the set. Review the details of each duplicate CI in the task and the data that was used to determine that the CI is a duplicate. For more information about IRE processes that detect duplicate CIs and generate de-duplication tasks, see [Detecting duplicate CIs](#).

Under some circumstances, IRE doesn't automatically generate de-duplication tasks for duplicate CIs. To still remediate the duplication in such case, you must create the de-duplication task manually. For information about when this happens and how to manually create a task, see [Manually create a de-duplication task](#).

Remediation of a de-duplication task consolidates its set of duplicate CIs into a single CI, eliminating the duplication.

To process de-duplication tasks (automatically-generated by IRE or manually created) to remediate duplicate CIs:

- In bulk: Use the De-duplication dashboard and the De-duplication Template Library in CMDB Workspace to:

- Review de-duplication tasks
- Remediate de-duplication tasks in bulk by using de-duplication templates
- Gain insights into de-duplication counts and status in your organization

Access the [CI de-duplication experience in CMDDB Workspace](#) to create de-duplication templates that remediate similar de-duplication tasks in bulk.

- A single task: [Review de-duplication tasks \(manual\)](#) and use the Duplicate CI Remediator wizard to process a single de-duplication task.

Main CI

The main CI plays a key role in the remediation of duplicate CIs. The main CI is one of the duplicate CIs that you want to retain as an active CI after reconciling the rest of the duplicate CIs into that main CI. You can then potentially retire or delete the rest of the duplicate CIs, or assign custom values to their attributes. The first step during remediation is to select a main CI for the remediation process. During remediation, you choose which attribute values, relationships, and related items from the duplicate CIs to reconcile into the main CI. Alternatively, you can choose not to consolidate any data and retain the main CI as it is.

The duplicate_of attribute in duplicate CIs, is used to store a reference to the main CI. For duplicate CIs which existed in an instance that was upgraded to the New York release or later, the main CI is unknown. After upgrade, duplicate_of for those duplicate CIs is set to 'Unknown', indicating that the CI is a duplicate but the main CI is unknown.

Before remediation, the CIs in a duplicate CIs set are all duplicates of each other. After remediation, a set of duplicate CIs consists of one main CI, and any number of CIs, each considered a duplicate of the main CI. The duplicate_of attribute of the main CI is empty. The duplicate_of attribute for all the rest of the duplicate CIs in the set, is a reference to the main CI of the set.

Default related items list

A list of related items that is used globally in the remediation of duplicate CIs, with all de-duplication tasks. During remediation, all items from the default related items list are selected by default to be merged to the

main CI. During remediation, adding or removing related items to be merged doesn't affect the default related items list. See [Manage default related items list](#) for more information.

Note:

- Merging of attributes and related items that are associated with assets is not supported. Asset related tables are not included in the default related items list and therefore they are not available for merge.
- If a scenario involves an inactive change request, the **Configuration item** field on the Change Request form is cleared. If the current value is a duplicate CI, then it isn't merged with the main CI.

De-duplication dashboard

[CMDB Workspace](#) provides a de-duplication remediation solution letting you remediate de-duplication tasks, in bulk. The De-duplication dashboard, and the de-duplication templates and libraries, let you remediate multiple de-duplication tasks, consistently, in a single operation. The De-duplication Template Library in CMDB Workspace lets you create a preconfigured de-duplication template with the remediation settings for a class. The De-duplication dashboard in CMDB Workspace lets you then run the preconfigured template on multiple de-duplication tasks for the class that is set in the template. The remediation settings in the template, are then consistently applied to the duplicate CIs of the de-duplication tasks, to remediate CI duplication.

For more information about remediating duplicate CIs in CMDB Workspace, see [CI de-duplication experience in CMDB Workspace](#).

Duplicate CI Remediator

A de-duplication remediation tool, that is wizard-like and that lets you reconcile a set of duplicate CIs associated with a single de-duplication task. You can choose one of the duplicate CIs to retain as an active CI, and then decide how to process the rest of the duplicate CIs. The Duplicate CI Remediator lets you set reconciliation options for attributes, relationships, and related items.

For information about using the Duplicate CI Remediator, see [Remediate a de-duplication task \(manual\)](#).

Effects on related tables (such as change requests)

Remediation can potentially have the following effects on change requests:

- When a CI is associated with a change request whose state isn't **New**, then when duplicate CI remediation processes delete the CI, the CI is removed from the change request.
- When a CI is associated with a change request whose state is **New**, then when duplicate CI remediation processes delete the CI, the CI in the change request is updated to the main CI.

The business rule Ready only CI when not New prevents change requests that aren't in the New state, from getting updated. This behavior ensures the validity and the continued processing of a change request when change request CIs are affected by duplicate remediation processing.

When a CI is associated with a problem or an incident task, those tasks remain valid after duplicate CI remediation processes run as the system merges the main CI in those tasks even if the remediation processes deleted the associated CI.

Properties that affect processing of duplicate CIs

During CMDB Identification, processing of sets of duplicate CIs is determined by:

- Property glide.identification_engine.skip_duplicates (true by default).
- Property glide.identification_engine.skip_duplicates.threshold (5 by default).
- Number of duplicate CIs in a set.

For information about how these properties affect the management of duplicate CIs, see [Detecting duplicate CIs](#).

Special remediation scenarios

There are a few special remediation scenarios in which the remediation of duplicate CIs behaves differently:

Large number of duplicate CIs

Support for reconciliation of duplicate CIs is limited when the number of duplicate CIs exceeds a certain threshold. This threshold is based on the value of the `glide.duplicate_ci_remediator.max.cis` property, which is 1,000 by default. You can update this property to increase the threshold. However, this threshold never exceeds 5,000, even if you set the property to a value greater than 5,000.

When the number of duplicate CIs for a de-duplication task exceeds the threshold, the options available in the wizard are limited:

- On the **Select Main CI** tab, only the **Recommended** list of main CIs appears, and only the **Use Main CI** option is available.
- Recommendations are based only on the oldest created, newest updated, and most recently discovered CIs.
- Reconciliation of attribute conflicts and CI relationships is not supported, and only default related items are reconciled.

Duplicate serial numbers

Duplicate CI remediation is usually applied to duplicate CMDB CIs. However, in some situations de-duplication tasks might be created for duplicate serial numbers. When remediation is applied to duplicate serial numbers, the merge of relationships from duplicate records, is not referenced and is not applied.

Large number of related items

A de-duplication task might be associated with a large number of related items which might block the remediation process as loading the task times out. To continue the remediation process in this case, you can try running remediation in a restricted mode. Add the `glide.duplicate_ci_remediator.enable_restricted_mode` system property, and set its value to **true** to show the **Use the Duplicate CI Remediator (Restricted Mode)** option in the Remediate dialog box. For more information about using this option to restrict the use of related items in de-duplication remediation and allow remediation to proceed, see [Using restricted mode within the Duplicate CI Remediator \[KB1542272\]](#).

Restrictions

IRE uses the `duplicate_of` field internally by populating it as part of the skip duplicate mechanism, and you should restrict manual updates of that field. For more details, see [Detecting duplicate CIs](#).

If you do attempt to modify the value of `duplicate_of` directly on a CI form or by using a script, the following restrictions are enforced to ensure data integrity:

- A CI cannot be its own main CI (you cannot set a CI as a duplicate of itself).
- A CI and its main CI cannot be from different domains.
-

The `duplicate_of` attribute of the main CI cannot reference any CI as its main CI (you cannot set a CI as a duplicate of another duplicate CI to create a chain of duplicate CIs).

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If you attempt to set a CI as a duplicate of another duplicate CI, then the CI is set as a duplicate of the main CI of the duplicate CI you are trying to set. If the main CI of the duplicate CI you are trying to set is 'Unknown', the operation fails.

Example: Attempt to set a CI as duplicate of another duplicate CI

CIs	Attempted setting	Result (System enforced)
CI1: <code>duplicate_of</code> = empty		CI1: <code>duplicate_of</code> = CI3
CI2: <code>duplicate_of</code> = CI3	CI1: <code>duplicate_of</code> = CI2	CI2: <code>duplicate_of</code> = CI3
CI3: Main CI		CI3: Main CI

If CI2 is a duplicate of 'Unknown', the operation fails.

If a main CI becomes a duplicate of another CI, then it can no longer be a main CI. All CIs that were duplicates of that main CI are set as duplicates of the new main CI.

Example: Attempt to set a main CI as duplicate of another CI

CIs	Attempted setting	Result (System enforced)
CI1: duplicate_of = CI4	CI4: duplicate_of = CI5	CI1: duplicate_of = CI5
CI2: duplicate_of = CI4		CI2: duplicate_of = CI5
CI3: duplicate_of = CI4		CI3: duplicate_of = CI5
CI4: Main CI		CI4: duplicate_of = CI5
CI5: duplicate_of = empty		CI5: Main CI

If a main CI becomes a duplicate of a CI within the same duplicate CI set, then the selected duplicate becomes the main CI in the duplicate CI set. The rest of the duplicate CIs in the set are set as duplicates of the new main CI.

Example: Attempt to set a main CI as duplicate of a CI within the duplicate CIs set

CIs	Attempted setting	Result (System enforced)
CI1: duplicate_of = CI4 CI2: duplicate_of = CI4 CI3: duplicate_of = CI4 CI4: Main CI	CI4: duplicate_of = CI1	CI1: Main CI CI2: duplicate_of = CI1 CI3: duplicate_of = CI1 CI4: duplicate_of = CI1

- You can't delete a CI that is the main CI for a set of duplicate CIs. To delete a main CI, you must first disassociate that main CI with all of its duplicate CIs. Either delete all duplicate CIs that are associated with that main CI, or remove the reference to that main CI from all duplicate_of attributes in any duplicate CIs that have it.
- [CI de-duplication experience in CMDB Workspace](#)
Create libraries of de-duplication templates in CMDB Workspace, to remediate de-duplication tasks in a consistent manner, and in bulk. De-duplication templates are pre-configured with all necessary settings for remediating duplicate CIs in de-duplication tasks, per class.
- [Review de-duplication tasks \(manual\)](#)
Review details of de-duplication tasks, and then potentially remediate a de-duplication task.
- [Remediate a de-duplication task \(manual\)](#)
Remediate a single de-duplication task by using the Duplicate CI Remediator wizard. Use the wizard to guide you through the duplicate CI reconciliation process or to apply a custom workflow.

- [Manually create a de-duplication task](#)

Manually create a de-duplication task when it is not automatically created. You can then use the Duplicate CI Remediator or the De-duplication dashboard in CMDB Workspace to remediate the manually created task.

- [Manage default related items list](#)

You can add or remove items from the default list of related items which is used in duplicate CI remediation for all de-duplication tasks.

- [Properties related to remediation of duplicate CIs](#)

Use de-duplication properties to configure how remediation of duplicate CIs works when using the Duplicate CI Remediator or the De-duplication dashboard in CMDB Workspace.

- [Components installed for duplicate CI remediation](#)

Tables installed to support duplicate CI remediation (included in the com.snc.cmdb plugin).

CI de-duplication experience in CMDB Workspace

Create libraries of de-duplication templates in CMDB Workspace, to remediate de-duplication tasks in a consistent manner, and in bulk. De-duplication templates are pre-configured with all necessary settings for remediating duplicate CIs in de-duplication tasks, per class.

Use the De-duplication dashboard in [CMDB Workspace](#) to see an overview and insights about duplicate CIs in your CMDB. The De-duplication dashboard lets you create de-duplication templates and organize these templates into libraries. You can also examine de-duplication tasks and then run a preconfigured template to remedy duplicate CIs in multiple de-duplication tasks in a single operation.

More information:

- To learn about general duplicate CI remediation concepts, restrictions, and special cases such as remediations that involve a large number of duplicate CIs, see [Duplicate CIs remediation](#).

- To remediate a single de-duplication task manually, using the Duplicate CI Remediator, see [Remediate a de-duplication task \(manual\)](#).
- To learn about Identification and Reconciliation (IRE) processes that detect duplicate CIs and other concepts related to duplication CI, see [Detecting duplicate CIs](#).

Access

To access the de-duplication functionality in the CMDB Workspace, navigate to **Workspaces > CMDB Workspace** and select **Management** in the CMDB Workspace menu bar. Then select the following links in Management tools, in the Manage section:

- **De-duplication Dashboard**
- **De-duplication Template Library**

Role requirement:

- To access the [Management view in CMDB Workspace](#) to perform de-duplication tasks, `sn_cmdb_admin`
- To perform de-duplication tasks, `cmdb_dedup_admin` or any role containing `cmdb_dedup_admin` (such as `sn_cmdb_admin`)

De-duplication templates and libraries

Use a de-duplication template to apply a consistent remediation to multiple, similar, de-duplication tasks, in a single action. A de-duplication template contains the settings that are required to remediate de-duplication tasks for a class. For example, how to choose the main CI, which method to use to consolidate attributes, and what to do with the duplicate CIs (aside from the main CI) after remediation. A de-duplication template also contains conditions that de-duplication tasks are matched on. When you run a de-duplication template, the remediation settings in the template are applied to all de-duplication tasks that match the template conditions.

To be able to run a de-duplication template, the template must be in a published state (a template is initially in a draft state). When published, you can create a schedule for a de-duplication template so it will automatically run on a recurring time interval. When necessary, you can unpublish a published template so it is temporarily unavailable for use until you republish it.

Organize your de-duplication templates into libraries to group templates by categories that are meaningful in your organization and to help locate a template. For example, you can create several de-duplication templates, all of which for the same CMDB class such as `cmdb_ci_win_server`. Then, add all those templates to the 'De-duplication templates for `cmdb_ci_win_server`' library for easy access.

De-duplication dashboard

The De-duplication dashboard in CMDB Workspace is a central location for remediating de-duplication tasks. Use the dashboard to:

- See all de-duplication templates on a set of cards. Each card shows details about a template. On a template card, you can select **Assign task** to assign de-duplication tasks to the template, and run the template to remediate its associated de-duplication tasks. You can also create a recurring schedule for a template such as daily and monthly.
- Select **Run all** to run all templates, each template remediating its assigned and matching de-duplication tasks.
- Select **View de-duplication tasks** to access a list view of de-duplication tasks with details such as the de-duplication template assignment. In the list view, you can select **Only show tasks that do NOT match any template conditions** to show only those de-duplication tasks that are unassigned or unmatched to any template class.
- Select **View template libraries** to create and manage de-duplication templates and libraries.
- Select the **Running** tab to track templates that are currently running.
- Select the **Completed** tab to check the status and details about the final state of a run, such as Completed, Cancelled, or Completed with errors.

De-duplication insights pane

The De-duplication insights pane on the right side of the De-duplication dashboard provides counts and insights related to the on-going de-duplication processes in your organization. This pane provides the following cards and links to help you resolve issues:

- Total count of de-duplication tasks that were remediated in the past week.
- De-duplication tasks that have completed with various types of errors and therefore need your attention. Provides the **View de-duplication tasks that need your attention** link to access a list view of the associated de-duplication tasks. You can examine error messages and drill down tasks for further troubleshooting details.
- De-duplication tasks that don't match the conditions in any template and therefore can't be assigned for remediation by a template. Provides the **View de-duplication tasks that don't match** link to access a list view of the associated de-duplication tasks. You can drill down tasks for further details.

Review the de-duplication templates and ensure that they're properly configured for remediating the on-going duplicate CIs that are being generated in the CMDB by newly added data.

Manage duplicate CIs skill

You can use the manage duplicate CIs skill that Now Assist for CMDB provides to get step-by-step guidance through the de-duplication process. The manage duplicate CIs skill guides you through reviewing de-duplication templates, assigning tasks to the templates, and then running the templates. The manage duplicate CIs skill presents you with relevant options for your next step, as you go, until you complete the remediation.

For more information, see [Now Assist for Configuration Management Database \(CMDB\)](#).

• [Create a de-duplication template](#)

Create a de-duplication template that you can then run to apply pre-configured and consistent remediation to de-duplication tasks, in bulk. Organize your templates in libraries that make sense in your organization.

• [Run a template to remediate de-duplication tasks](#)

Run a de-duplication template to remediate de-duplication tasks in bulk. A de-duplication template is pre-configured with remediation settings per class. Running a de-duplication template, remediates its

assigned de-duplication tasks, consistently, according to the template settings.

- [Remediate a single de-duplication task](#)

Remediate a single de-duplication task using a de-duplication template in CMDB Workspace, or manually, using the Duplicate CI Remediator.

- [Publish or unpublish a de-duplication template](#)

After creating a de-duplication template, you must publish the draft template to make it available to run. You might also need to unpublish a published de-duplication template so it's temporarily unavailable for de-duplication remediation, until you republish it again.

- [Schedule a de-duplication template](#)

Create a schedule, such as a monthly schedule, for a published de-duplication template so it will automatically run on a recurring interval.

- [Retire a de-duplication template](#)

Retire a template that is no longer needed for remediating de-duplication tasks.

Create a de-duplication template

Create a de-duplication template that you can then run to apply pre-configured and consistent remediation to de-duplication tasks, in bulk. Organize your templates in libraries that make sense in your organization.

Before you begin

Role required:

- To access the [Management view in CMDB Workspace](#) to perform de-duplication tasks, `sn_cmdb_admin`
- To perform this task, `cmdb_dedup_admin` or any role containing `cmdb_dedup_admin` (such as `sn_cmdb_admin`)

About this task

By default, a newly created template is in draft mode. Therefore, after creating a template, you must publish the template to make it available for remediation.

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
2. Select the **De-duplication Template Library** link in Management tools, in the Manage section.
3. (Optional) Create a library for the new template.
 - a. On the De-duplication templates page, select **Create a library**.
 - b. In the Create new library dialog box, fill out the **Name** and the **Description** for the new library.
 - c. Select **Create**.
4. On the De-duplication templates page, select **Create a template**. On the New template page, configure the different sections of the template by completing the rest of the steps.
5. **General information**: Fill out the form and then select **Next**.

Field	Description
Name	Unique name for the template.
Class	CMDB class that this template can be applied to.
Template library	Library that the de-duplication template belongs to. Select a newly created library or any other library. Select a library, or select the '+' icon to create a new library.

Option	Description
Automatically select all de-duplication tasks that match the selected class	Assign all de-duplication tasks that match the template class, to the template for remediation.
Add all child classes	Also assign all de-duplication tasks that match descending classes of the template class, to the template for remediation. This option is available only if the Automatically select all de-duplication tasks that match the selected class option is selected.

6. **Create condition:** Use a condition builder or a custom script to narrow down the list of de-duplication tasks assigned to the template, and then select **Next**.

Use script toggle options:

-

Off: Specify conditions to filter on CIs from the template class. Only the de-duplication tasks containing CIs that match the filter will be assigned to the template.

-

On: Enter a custom script that returns a valid array of de-duplication tasks sys_ids (from the Remediate Duplicate Task [reconcile_duplicate_task] table), that are applicable to the template class. Those de-duplication tasks will then be assigned to the template.

Note: Ensure that the script runtime doesn't exceed your intended runtime expectation and doesn't unexpectedly affect performance. Invalid values returned by the script are ignored.

7. **Select main CI:** Select the method for choosing the main CI from the drop-down list and then select **Next**.

Option	Description
Newest updated	Most recently updated CI.
Most related items	CI with most related items.
Oldest created	Oldest CI.
Most relationships	CI with most relationships.
Discovery source	A duplicate CI must be associated with the selected discovery source to become the main CI.
Has asset	A CI must have an asset to become the main CI.
Select main CI based on condition	A duplicate CI must match a specified condition to become the main CI.

Note:

- Applying the chosen method to a set of duplicate CIs must result in exactly one CI. If during remediation, the chosen method yields no CIs or multiple CIs, then the **Newest updated** method is automatically applied regardless of your choice.
- After selecting the main CI out of the set of duplicate CIs, the main CI is separate from the rest of the duplicate CIs. From that point, 'duplicate CIs' refers to all of the initial duplicate CIs, aside from the selected main CI.

8. **Merge attributes:** Configure merging attributes from duplicate CIs into the main CI, and then select **Next**.

- a. Select the method to use for merging the attributes of duplicate CIs into the corresponding attributes of the main CI.

For each method card on the left (such as Use latest updated CI), select attributes in the list on the right, that should be merged into the main CI using that method.

- b. Select options as appropriate.

Option	Description
Allow null value updates when merging attributes to main CI Allow null value updates from duplicate CIs to main CI	When merging duplicate CIs into the main CI, lets a null value from a duplicate CI to overwrite a non-null value in main CI.
Allow updates from duplicate CIs only when the main CI value is null	When merging duplicate CIs into the main CI: - Prevents updates to main CI attributes that have non-null value. - Allows updates to main CI attributes that have a null value, with a non-null value.

9. **Merge relationships:** Select an option for merging relationships from duplicate CIs into the main CI, and then select **Next**.

Choice	Description
Merge all relationships	Merge relationships from all duplicate CIs into the main CI.
Do not merge any relationships	Don't merge any relationships from duplicate CIs into the main CI.
Merge relationships based on condition	Merge relationships only if the main CI satisfies this condition.

10. **Merge related items:** Select the method to use for merging related items associated with the duplicate CIs into the main CI, and then select **Next**.

Choice	Description
Merge default related items	Merge only related items in the default related items list, from duplicate CIs into the main CI.
Do not merge any related items	Don't merge related items from any duplicate CI into the main CI.
Merge all available related items	Merge all related items from all duplicate CIs into the main CI.
Merge related items based on condition	Merge related items only if the main CI satisfies this condition. Related items are merged individually based on the main CI satisfying their respective condition. If there are two related items with two conditions and the main CI satisfies only one condition, then only one related item is merged. Select Create a new condition to build conditions.

For information about the default list of related items used in de-duplication processes, see [Manage default related items list](#).

11. **Duplicate CI actions:** Select what to do with the duplicate CIs after they're merged into the main CI, according to the template settings, then select **Next**.

Choice	Description
Delete	Permanently delete all duplicate CIs in the set (main CI remains the only active CI).

Choice	Description
Set attributes to custom values	Set attributes of duplicate CIs to custom values. Configure the pairs of attribute/value with your custom settings.

12. **Review and test:** Access the following tabs to complete the template by reviewing and testing it.
 - a. **Summary:** Review all template settings and select **Back** to modify any settings.
 - b. **Test:** Select a sample de-duplication task and then select **Save and Preview Results** to save and run the template.
 - c. Preview the results of the remediation test.
 - d. Select **Save & Exit** to save the template and return to the De-duplication templates home page.

Result

The new template is added to the specified library in draft mode and can be edited. However, a template in draft mode isn't available for remediation of de-duplication tasks until you publish it.

What to do next

- [Publish a de-duplication template](#)
- [Run a template to remediate de-duplication tasks](#)
- [Remediate a single de-duplication task](#)

Run a template to remediate de-duplication tasks

Run a de-duplication template to remediate de-duplication tasks in bulk. A de-duplication template is pre-configured with remediation settings per class. Running a de-duplication template, remediates its assigned de-duplication tasks, consistently, according to the template settings.

Before you begin

Role required:

- To access the [Management view in CMDB Workspace](#) to perform de-duplication tasks, `sn_cmdb_admin`
- To perform this task, `cmdb_dedup_admin` or any role containing `cmdb_dedup_admin` (such as `sn_cmdb_admin`)

About this task

You can run a single template to remediate its associated de-duplication tasks, or run all published templates at once, each remediating duplicate CIs in its associated tasks. Before running any templates, ensure that those templates are properly configured with their remediation settings and are properly assigned with de-duplication tasks. De-duplication tasks can be automatically assigned to a template based on the template conditions. You can also manually assign additional de-duplication tasks to a template.

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
2. Select the **De-duplication Dashboard** link in Management tools, in the Manage section.
3. Select a template library in the **Ready** tab, or select **All libraries** in the libraries drop-down list (set to **All libraries** by default).
The set of cards is filtered to show only the templates from the selected library, or to show all templates from all libraries if you select **All libraries** from the drop-down list.
4. (Optional) On the De-duplication templates page, select **Run all** to run all de-duplication templates on their assigned de-duplication tasks.
To assign de-duplication tasks to a template or to run a single template, continue with the following steps.
5. On the De-duplication templates page, locate the card with the template that you want to run and select **Assign tasks** in that card.

6. Select the tasks in the Available de-duplication tasks list that you want to remediate using the selected de-duplication template. Then select **Add Selected** to assign the selected tasks to the current template.

Available de-duplication tasks lists all open de-duplication tasks, some of which can be non-compatible with the template. If you try to associate a non-compatible task, an error appears. Some of the tasks in the list might be assigned to another template, and you can reassign those to the current template.

The Assigned de-duplication tasks pane on the right contains two tabs:

- **All matching tasks:** Shows de-duplication tasks that match the condition of the selected template.
- **Your selected tasks:** Shows the tasks that you've assigning to the template for remediation.

The selected tasks are added to the list in the **Your selected tasks** tab in the Assigned de-duplication tasks pane on the right. The template will run for all tasks in the **Your selected tasks** list and any tasks in the **All matching tasks** list that aren't assigned to another template.

7. Select **Run template**.
The selected template is added to the running queue.

What to do next

- Select the **Running** tab to track the progress and details of templates that you added to the running queue and are currently running. General counts for the template run appear when those counts become available. The templates are sorted by their start time, with the template with the most recent start time at the top of the list.
- Select **Details** (accessible only after all de-duplication tasks have been calculated) to show further details about this run and to drill down to de-duplication tasks.
- Select **Cancel** to stop a template's run. The details for the cancelled template are added in the **Completed** tab with the state of Cancel Requested. Upon completing the cancellation, the state changes to Cancelled.

After a template run completes, its details no longer appear under the **Running** tab, but are available in the **Completed** tab. Also, the completed template is available again for running in the **Ready** tab.

- Select the **Completed** tab to view general counts per template, such as erroneous and skipped tasks, and general run time. The templates are sorted by their completion time, with the template that completed most recently at the top of the list. Select **Details** for a template to access a list view of the template's assigned de-duplication tasks. This list view shows details for each de-duplication task, such as its state (color coded) and error message if applicable. You can drill down a de-duplication task for further details.
- Select **View de-duplication tasks that need your attention** in the De-duplication tasks completed with errors card in the De-duplication insights pane on the right. The De-duplication tasks with errors list view shows de-duplication tasks that didn't complete successfully along with an error message. You can drill down de-duplication tasks in the list for more details.
- [Schedule a de-duplication template.](#)

Remediate a single de-duplication task

Remediate a single de-duplication task using a de-duplication template in CMDB Workspace, or manually, using the Duplicate CI Remediator.

Before you begin

Role required:

- To access the [Management view in CMDB Workspace](#) to perform de-duplication tasks, sn_cmdb_admin
- To perform this task, cmdb_dedup_admin or any role containing cmdb_dedup_admin (such as sn_cmdb_admin)

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
2. Select the **De-duplication Dashboard** link in Management tools, in the Manage section.

3. In the De-duplication templates section, select **View de-duplication tasks**.
4. In the De-duplication tasks list view, select the task that you want to remediate.
5. Select **Remediate** on the task form.
6. In the Remediate dialog box, choose which method you want to use for remediation, and then select **Remediate**.

Choice	Description
Remediate manuallyUse the Duplicate CI Remediator	Use the Duplicate CI Remediator built on Core UI to remediate the task. To continue with this choice of remediation, see Remediate a de-duplication task (manual).
Use the Duplicate CI Remediator in Restricted Mode	Appears only if the <code>glide.duplicate_ci_remediator.enable_restricted_mode</code> system property is set to true. You might need to use this option if remediation is blocked because loading the de-duplication task times out. This option might be relevant when there is a large number of related items associated with a de-duplication task, allowing remediation to continue with the limited features in a restricted mode. For more information about using this option to restrict the use of related items in de-duplication remediation and allow remediation to proceed, see Using restricted mode within the

Choice	Description
	Duplicate CI Remediator [KB1542272] . To continue with this choice, see Remediate a de-duplication task (manual) . Skip to step #5 in the procedure.
Remediate using a template	Use a de-duplication template to remediate the task. To continue with this choice of remediation: In the Remediate dialog box, select the Library and the Template to use for the remediation, and then select Remediate .

What to do next

On the task form:

- Track the progress and the details of remediation in the Activity stream until remediation is complete.
- Select the **Duplicate Audit Results** tab to see the results of the duplicate audit.

Publish or unpublish a de-duplication template

After creating a de-duplication template, you must publish the draft template to make it available to run. You might also need to unpublish a published de-duplication template so it's temporarily unavailable for de-duplication remediation, until you republish it again.

Before you begin

- A template that you want to publish must not be in a published state.

- A template that you want to unpublish must be currently published.
- If the template that you want to unpublish is currently running or in the running queue, you must wait for the run to complete or cancel the run. In the De-duplication dashboard, select the **Running** tab, locate the template that you want to unpublish, and then select **Cancel** in the template's card.

Role required:

- To access the [Management view in CMDB Workspace](#) to perform de-duplication tasks, `sn_cmdb_admin`
- To perform this task, `cmdb_dedup_admin` or any role containing `cmdb_dedup_admin` (such as `sn_cmdb_admin`)

About this task

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
2. Select the **De-duplication Template Library** link in Management tools, in the Manage section.
3. In the libraries pane locate the template that you want to publish or unpublish and then either:
 - a. Publish: Ensure that all the sections of the template are complete and select **Publish**.
 - b. In the Publish template dialog box, select **Publish**.
 - c. Unpublish: Select **Unpublish** and in the Unpublish template dialog box, select **Unpublish**.

Result

A published template:

- Becomes available on the De-duplication templates page and can be used for remediation.
- Can't be edited or retired.

An unpublished template is set back into draft mode:

- Is no longer available on the De-duplication templates page and can't be selected for remediation.
- Can be edited or retired.

Schedule a de-duplication template

Create a schedule, such as a monthly schedule, for a published de-duplication template so it will automatically run on a recurring interval.

Before you begin

The de-duplication template that you want to create a schedule for, must be in a published state. For information about publishing a template, see [Publish or unpublish a de-duplication template](#).

Role required:

- To access the [Management view in CMDB Workspace](#) to perform de-duplication tasks, sn_cmdb_admin
- To perform this task, cmdb_dedup_admin or any role containing cmdb_dedup_admin (such as sn_cmdb_admin)

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
2. Select the **De-duplication Dashboard** link in Management tools, in the Manage section.
3. On the De-duplication templates pane, select the More options icon  () for the template that you want to create a schedule, and then select **Create schedule**.
4. On the Schedule <template> dialog box, fill in the scheduling details. Various fields, such as **Time zone** and **Time** appear on the form according to your choice in the **Run** field. If **Run** is **Periodically**, you

must set the **Repeat interval (day)** or the **Repeat interval (time)** fields (or both) before you can save the schedule.

5. Select **Save**.

Result

The newly created schedule is active by default.

What to do next

Track the state and details of templates:

- Select the **Running** tab to track the progress and details of the scheduled templates that are currently running. General counts for the template run appear when those counts become available. The templates are sorted by their start time, with the template with the most recent start time at the top of the list.
- Select **Details** (accessible only after all de-duplication tasks have been calculated) to show further details about this run and to drill down to de-duplication tasks.
- Select **Cancel** to stop a template's run. The details for the cancelled template are added in the **Completed** tab with the state of Cancel Requested. Upon completing the cancellation, the state changes to Cancelled.

After a template run completes, its details no longer appear under the **Running** tab, but are available in the **Completed** tab. Also, the completed template is available again for running in the **Ready** tab.

- Select the **Completed** tab to view general counts per template, such as erroneous and skipped tasks, and general run time. The templates are sorted by their completion time, with the template that completed most recently at the top of the list. Select **Details** for a template to access a list view of the template's assigned de-duplication tasks. This list view shows details for each de-duplication task, such as its state (color coded) and error message if applicable. You can drill down a de-duplication task for further details.
- Select **View de-duplication tasks that need your attention** in the De-duplication tasks completed with errors card in the De-duplication

insights pane on the right. The De-duplication tasks with errors list view shows de-duplication tasks that didn't complete successfully along with an error message. You can drill down de-duplication tasks in the list for more details.

On the De-duplication templates pane, select the More options icon



() for a template with a schedule. Some of the following options appear, according to the state of the schedule:

- Select **Edit schedule** to modify the schedule.
- Select **Deactivate schedule** to halt temporarily the automatic runs of the template, while preserving the schedule settings.
- Select **Activate schedule** to resume automatic runs of the template according to the schedule settings.

Retire a de-duplication template

Retire a template that is no longer needed for remediating de-duplication tasks.

Before you begin

The template that you want to retire must be in an [unpublished state](#).

Role required:

- To access the [Management view in CMDB Workspace](#) to perform de-duplication tasks, `sn_cmdb_admin`
- To perform this task, `cmdb_dedup_admin` or any role containing `cmdb_dedup_admin` (such as `sn_cmdb_admin`)

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
2. Select the **De-duplication Template Library** link in Management tools, in the Manage section.

3. In the library pane, select the template that you want to retire and then select **Retire**.
4. Select the template's context menu and then select **Retire template**.
5. In the Retire template dialog box, select **Retire**.

Result

A retired template is no longer available in the De-duplication dashboard and can't be used for remediation.

Review de-duplication tasks (manual)

Review details of de-duplication tasks, and then potentially remediate a de-duplication task.

Before you begin

Role required: itil to view, and itil_admin or cmdb_dedup_admin to remediate a de-duplication task.

About this task

Alternatively, you can use de-duplication capabilities in the CMDB Workspace. Use the De-duplication dashboard and the De-duplication Template Library to:

- Review de-duplication tasks
- Remediate de-duplication tasks in bulk by using de-duplication templates
- Gain insights into de-duplication counts and status in your organization

For more information, see [CI de-duplication experience in CMDB Workspace](#).

If a duplicate CI is a dependent CI, then you can view the details of the dependent relationship, the Depend on CI, and any relation qualifier chain. If the dependent CI has a lookup table, then you can see the details of the respective lookup table.

Procedure

1. Navigate to **All > Configuration > Identification/Reconciliation > De-duplication Tasks**.
2. Select a task.

Remediate Duplicate Task form

Field	Description
Number	Unique task number.
Assigned to	Person who is responsible for resolving the task.
Short description	Description for the task.
Work notes	Details describing how the CI was identified as a duplicate. Note: Not available in de-duplication tasks that were created prior to the London release. This field also contains user notes about the decisions and steps of resolving the task.
Priority	Task priority.
State	State of the de-duplication task as it progresses through resolution.

3. In the related lists section, click the **Duplicate Audit Results** tab to see the list of duplicate CIs in this task. You can click a CI to display more CI details.

Column	Description
Duplicate CI	Reference to the duplicate CI. Note: This field is a document ID type, which means that it can reference any record on any table. If the referenced CI is deleted as part of resolving duplicate tasks, then this field is empty.
Relationship	For a duplicate CI that is a dependent CI, this field shows the relationship between the duplicate CI and depend on CI.
Depend on CI	If the duplicate CI is a dependent CI, then this field displays the depend on CI.
Discovery source	Discovery method used for the CI.

What to do next

Analyze de-duplication tasks to determine which CIs should remain active and which of the duplicate CIs in the Duplicate Audit Results lists are stale or incorrect. Click **Remediate** to [remediate a single de-duplication task](#).

Related concepts

- [Relation qualifier](#)

Related topics

- [Document ID field](#)

Remediate a de-duplication task (manual)

Remediate a single de-duplication task by using the Duplicate CI Remediator wizard. Use the wizard to guide you through the duplicate CI reconciliation process or to apply a custom workflow.

Before you begin

Review the following topics to familiarize yourself with important concepts of duplicate CI remediation:

- [Duplicate CIs remediation](#) to learn about general duplicate CI remediation concepts, restrictions, and special cases such as remediations that involve a large number of duplicate CIs.
- [Properties related to remediation of duplicate CIs](#) for information about important properties that affect processes of duplicate CI remediation. Including the `glide.duplicate_ci_remediator.dry_run` property that determines if the Duplicate CI Remediator actually updates the CMDB or not.

Role required: `itil` to read, `itil_admin` or `cmdb_dedup_admin` to write

About this task

Instead of using the build of the Duplicate CI Remediator on Core UI, you can use the De-duplication dashboard, and de-duplication templates and libraries, available in [CMDB Workspace](#). Using de-duplication capabilities in CMDB Workspace is especially useful for bulk remediation of de-duplication tasks. These capabilities let you pre-configure a de-duplication template with remediation settings, which can then be applied in a consistent manner to multiple de-duplication tasks. For more information, see [CI de-duplication experience in CMDB Workspace](#).

As you progress through the tabs of the Duplicate CI Remediator, CIs are not updated. All updates are applied only in the final step, after you click **Remediate**.

Note: Remediation behaves differently when remediation involves a large number of duplicate CIs or where the duplicates are serial numbers. For information about special cases in remediation, see [Duplicate CIs remediation](#).

Procedure

1. Navigate to **All > Configuration > Identification/Reconciliation > De-duplication Tasks**.
2. Open the de-duplication task that you want to remediate.
3. On the task form, select **Remediate**.
4. In the Remediate dialog box, select either of the following options:

- **Use the Duplicate CI Remediator (Recommended):** Use the wizard to consolidate duplicate CIs according to your configurations and settings. Follow the Duplicate CI Remediator tabs to configure the reconciliation.

Note: This option is not available with non-CMDB tables.

- **Use a custom remediation workflow:** Use an existing CMDB remediation rule or select **Add New** to [create a new one](#).
 - On the CMDB Remediation Rule form, set **Task type** to **Remediate Duplicate Task** and select **Active**.
 - On the Workflow form, set **Table** to **Remediate Duplicate Task** [reconcile_duplicate_task] and **If condition matches** to **None**.
 - Ensure that the associated workflow remediates duplicate CIs.

In the Remediate dialog box, select **Next** to start the workflow and to exit the Duplicate CI Remediator. The Remediate Duplicate Task form appears, where you can update the **State** of the task.

- **Use the Duplicate CI Remediator (Restricted Mode):** Appears only if the glide.duplicate_ci_remediator.enable_restricted_mode system property is set to **true**. You might need to use this option if remediation is blocked because loading the de-duplication task times out. This option might be relevant when there is a large number of related items associated with a de-duplication task, allowing remediation to continue with the limited features in a restricted mode. For more information about using this option to restrict the use of related items in de-duplication remediation and allow remediation to proceed, see [Using restricted mode within the Duplicate CI Remediator \[KB1542272\]](#).

5. On the **Select Main CI** tab in the Duplicate CI Remediator:
 - a. Select the main CI for this reconciliation using either of the following lists of duplicate CIs. For any CI, you can click the **Name** link to display the CI's attributes, or click the **Related Items** link to display the number of related items.
 - **Recommended:** A subset of the **All** list, containing only system recommended main CIs. System recommendations are based on checking the duplicate CIs for the following criteria:
 - CI with most related items.
 - CI with most relationships.
 - Newest discovered CI.
 - Newest updated CI.
 - Oldest created CI.
 - Previous main CI, if one was previously selected.
 - **All:** All duplicate CIs for the de-duplication task.

Either option lets you review a summary of the remediation settings before starting the remediation.

- b. Select one of the following options to choose whether to consolidate any attribute values, relationships, or related items from any of the duplicate CIs into the main CI:
 - **Remediate Manually:** Lets you specify which attribute values, relationships, and related items from duplicate CIs to consolidate into the main CI.
 - **Use Main CI:** Retains main CI attribute values, merges relationships, and merges only the default related items.

Skip to step number 7 as this selection skips all configurations other than choosing the action for the duplicate CIs on the **Determine Duplicate CI Actions** page.

6. On the **Merge Attribute Values** tab review inconsistent values of each attribute. For each attribute, choose to retain the main CI's value, or choose a value from a duplicate CI for the main CI to be set with.
7. Select **Next**.

Column	Description
Attribute	Attribute for which there are different values.
Main CI Value	Attribute value in the main CI.
Other Values	The number of unique Attribute values within the duplicate CIs.

To override the main CI attribute value with a duplicate CI value:

- a. Click the **Other Values** link.
- b. In the attribute dialog box, click **Unique Values** to display only unique attribute values, or **All** to display all attribute values including identical values.
- c. Select a value for the main CI **Attribute** to be set with.
You can click **Reset to Original** to undo the selection of a different attribute value for the main CI.
- d. Click **Select**.

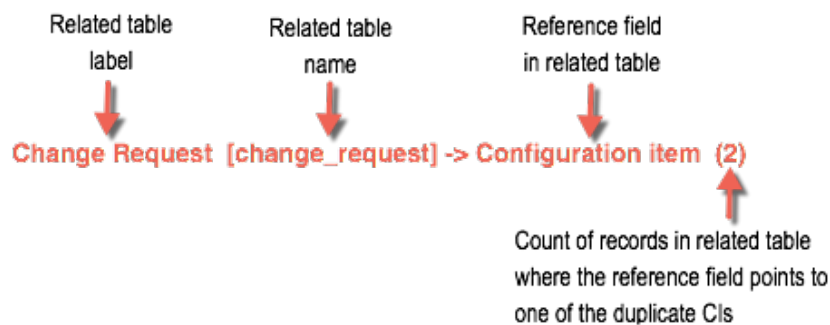
Note: Attributes, such as system fields, discovery fields (discovery_source, last_discovered, first_discovered), and read-only fields (such as the asset field) do not appear in the list.

8. On the **Merge Relationships and Related Items** tab:
 - a. In the Merge Relationships section, select whether to merge all relationships from all duplicate CIs into the main CI. Click **View all relationships** to display all the relationships in which a duplicate CI is a parent or a child. You can click a **Parent** or a **Child** link to display more details. Orphan and duplicate relationships of duplicate CIs are deleted if you choose to merge relationships.

For more information about the CMDB Health relationship KPI, see [CMDB Health KPIs and metrics](#).

- b. In the Merge Related Items section use the list collector to select related items to be merged into the main CI. Click **View all related items** to display all related tables (items) and the count of references in each table to one of the duplicate CIs. You can click the links in **Main CI Related Items** and **Duplicate CIs Related Items** to display details about the related items.

Related items in the list collector have the following format:



- By default, all items in the default related items list are selected to be merged.
- Related items (tables) that have no references to a duplicate CI are not listed, unless that table is included in the default related items list.
- Since asset related tables are part of the exclusion list, they are not available for merge.

See [Manage default related items list](#) for more information about configuring a default list of related items.

- c. Click **Next**.
9. On the **Determine Duplicate CI Actions** tab, choose one of the following actions to perform after completing the reconciliation. Then click **Next**.
 - **Set attributes to custom values** (recommended): Retain all duplicate CIs. Mark the duplicate CIs as invalid by setting

a specific **Attribute** to a specific **Value** for all duplicate CIs. For example, set Operational Status to **Retired** to retire the duplicate CIs.

The duplicate_of attribute of the duplicate CIs is automatically set to the appropriate main CI. Also, the duplicate CIs will not be added to any de-duplication task after this task is remediated. If the identification engine is not configured to [skip duplication](#), ensure that identification inclusion rules are configured to exclude the duplicate CIs during identification. This configuration prevents new de-duplication tasks with the same duplicate CIs from being created after remediation.

Note: Discovery fields, system fields, read only fields, and date fields are excluded from the attributes list.

- **Delete:** Delete all duplicate CIs (only the main CI remains).

Note: Review [Roll back and delete recovery](#) for information about reverting the deletion of CIs and related records.

10. On the **Review and Confirm** tab:

- Review the summary of the expected updates for this duplicate CIs reconciliation. Updates are based on your selections and therefore the summary includes only the details that are applicable. This summary can include details of the relationships and related items that will be merged to the main CI, the attribute values that the main CI will be set with, and the number of CIs that will be deleted. Click **Attributes**, **Relationships**, **Related Items**, or **Duplicate CI Actions** if applicable, to display further details such as changes to attribute values.
- Click **Remediate** to complete the reconciliation according to your reconciliation settings.
Once complete, the task **State** is set to **Closed Complete**.

Result

The following relationships are deleted without being merged to the main CI:

- Relationships in which the type, child, or parent field is empty.

- Relationships for which merging to the main CI will result in cyclic relationships.
- Relationship for which merging to the main CI will result in duplicate relationships.

What to do next

The reconciliation process runs in the background and may take a while to complete. Upon completion, the system sends a confirmation notice to the remediator of the task. Meanwhile, you can:

- **Review Identification Rules:** Review identification and inclusion rules and make any necessary updates to reduce CI duplication.
- **Check Progress:** View the task activities that are logged as the remediation progresses.
- **View Main CI:** View the main CI for this reconciliation process.

Manually create a de-duplication task

Manually create a de-duplication task when it is not automatically created. You can then use the Duplicate CI Remediator or the De-duplication dashboard in CMDB Workspace to remediate the manually created task.

In some situations, duplicate CIs are not automatically detected and de-duplication tasks are not automatically generated. Such situation happens with a class for which identification rules are not defined and the identification engine cannot be applied. However, you still want to reconcile these duplicate CIs by utilizing the [Duplicate CI Remediator](#) or the [De-duplication dashboard](#) in CMDB Workspace.

Use the [CMDBDuplicateTaskUtils](#) API to manually create a de-duplication task in which all duplicate CIs are specified. The de-duplication tasks that you create manually and the automatically created tasks, are stored in the same table ([reconcile_duplicate_task]) and are processed in the same manner.

Note: You can manually create a de-duplication task only for CMDB CIs and a CI can be specified as a duplicate CI only in a single de-duplication task.

Manage default related items list

You can add or remove items from the default list of related items which is used in duplicate CI remediation for all de-duplication tasks.

Before you begin

Role required: itil_admin

About this task

The default related items list is used by the Duplicate CI Remediator, appearing on the **Merge Relationships and Related Items** tab. You can modify the list for only a specific task in the Duplicate CI Remediator without affecting the default global list.

The default related items list is also used by the [De-duplication dashboard](#) in CMDB Workspace for remediation.

Globally modifying the list affects the remediation of all de-duplication tasks in the Duplicate CI Remediator and in the De-duplication dashboard in CMDB Workspace.

Note: Only related items in which the reference field points to Configuration Item [cmdb_ci] in sys_dictionary can be selected for the default related items list. Related items with references to any child of the Configuration Item class cannot be selected for the default related items list, but are still available for merging in the Duplicate CI Remediator for a specific task.

Procedure

1. Navigate to **All > Configuration > Identification/Reconciliation > Duplicate CI Remediator Default Related Items**.
2. On the Default Related Items List for Duplicate CI Remediator page, use the list collector to add or remove items from the **Selected** list.

3. Click **Save**.

Properties related to remediation of duplicate CIs

Use de-duplication properties to configure how remediation of duplicate CIs works when using the Duplicate CI Remediator or the De-duplication dashboard in CMDB Workspace.

To view and edit these properties, the admin role is required.

Although the properties below reference the [Duplicate CI Remediator](#), these properties have an equivalent effect in [duplicate CI remediation in CMDB Workspace](#).

Property	Description
Attributes in which max_length exceeds this property value (4000 by default) are excluded from the Select Main CI, Merge Attribute Values, and Determine Duplicate CI Actions tabs in the Duplicate CI Remediator wizard. glide.duplicate_ci_remediator.max_field_length	If the max_length for an attribute is equal to the property value, and the size of the data exceeds the property value, then the data is truncated to the property value and the attribute appears in attribute lists. - Type: integer - Default value: 4000 - Location: Configuration > CMDB Properties > Duplicate CI Remediator Properties Note: This property impacts the performance of de-duplication tasks, therefore be cautious about setting this value.
Comma separated list of related tables in the format	Type: string

Property	Description
'<table>.<reference column>', that are excluded from merging during duplicate CI remediation. glide.duplicate_ci_remediator.related_items_blacklist	- Default value: cert_task.cmdb_ci,cert_audit_result.configuration_item,discovery_log.cmdb_ci,alm_hardware.ci,alm_asset.ci,fm_expense_line.ci - Location: Configuration > CMDB Properties > Duplicate CI Remediator Properties
Threshold for the number of duplicate CIs, which if exceeded, support for reconciliation in the Duplicate CI Remediator is limited (1,000 by default). glide.duplicate_ci_remediator.max_cis	- Type: integer - Default value: 1000 - Location: Configuration > CMDB Properties > Duplicate CI Remediator Properties - Learn more: See 'Large number of duplicate CIs' in Remediate a de-duplication task (manual). This threshold never exceeds 5,000, even if you set the property to a value greater than 5,000.
Determines whether the Duplicate CI Remediator actually remediates CI duplication by updating records in the CMDB, or not. glide.duplicate_ci_remediator.dry_run	When set to false (default value), updates specified in the wizard are actually performed. You can set this property to true and then test run through the Duplicate CI Remediator without any records actually being updated. In this case, the work notes for the task describe the changes that will happen in an actual remediation. Type: true false

Property	Description
	• Default value: false • Location: Configuration > CMDB Properties > Duplicate CI Remediator Properties
glide.duplicate_ci_remediator.enable_restricted_mode	Enables restricted mode remediation in the Duplicate CI Remediator. This option restricts the use of related items in de-duplication remediation and allows a blocked remediation to proceed using the Use the Duplicate CI Remediator (Restricted Mode) option. For more information, see Using restricted mode within the Duplicate CI Remediator [KB1542272]. • Type: true false • Default value: false • Location: Add to System Properties [sys_properties] • Learn more: Remediate a de-duplication task (manual) Set to true to enable restricted mode remediation.

Components installed for duplicate CI remediation

Tables installed to support duplicate CI remediation (included in the com.snc.cmdb plugin).

Tables installed

Table	Description
Remediate Duplicate Task [reconcile_duplicate_task]	De-duplication tasks.
Duplicate Audit Result [duplicate_audit_result]	CIs associated with each de-duplication task.
Duplicate CI Remediation [cmdb_duplicate_ci_remediation]	Input, selections on each tab, overall status of remediation, and results of each run of the Duplicate CI Remediator.
CMDB Workspace De-duplication template library [sn_cmdb_ws_reconcile_duplicate_template_library]	Library names and their descriptions.
De-duplication template suggested task [sn_cmdb_ws_reconcile_duplicate_template_suggested_task]	Mappings of de-duplication tasks to templates.

Scheduled jobs installed

Scheduled job	Description
De-duplication: Populate Duplicate Task Data	Runs daily to copy metadata, such as the identification attributes and identification rules used in generating a de-duplication task,

Scheduled job	Description
	into an easily accessible table. The metadata is stored in the Reconcile Duplicate Task Data [reconcile_duplicate_task_data] table and used for analytics for the De-duplication dashboard.
De-duplication: Populate Template Task Run Records	Runs every minute to evaluate the de-duplication tasks applicable for a template according to your settings. Based on settings, assigns matching tasks and manually added tasks to the template for remediation. De-duplication tasks are remediated after you select Run template.
De-Duplication: Template Scheduler	Runs every minute by the template scheduler to check the capacity of the execution engine. If there is available capacity, picks any new de-duplication tasks added to the queue, for remediation. De-duplication tasks aren't remediated and remain in ready state unless this job runs periodically.
De-duplication: Populate Template Suggested and Orphan Tasks	Runs hourly to count the number of tasks that match each published template and the number of tasks that don't match any published template. These counts then

Scheduled job	Description
	appear on various cards in the De-duplication dashboard.

CIs attestation

Verify the existence of actual IT infrastructure and applications that you own, systematically and in bulk. As CIs are continuously ingested into the CMDB from various data sources, ensure the integrity of the CMDB. Remove any stale CIs that are associated with IT infrastructure or applications that no longer exists.

Use the [CMDB Data Manager in CMDB Workspace](#) to [create an Attestation policy](#), specifying CIs that need to be attested and the attestation frequency. Assign Attestation tasks to users that are familiar with or that manage the CIs, and who can attest or reject the IT infrastructure or applications that those CIs represent. Rejected CIs that are no longer needed can then be retired, archived, or deleted from the CMDB.

Users can go to the [My Work view in CMDB Workspace](#) to see their assigned attestation tasks, and then [review and process the tasks](#).

Smart detection and auto-attestation

Smart detection streamlines and simplifies CI attestation. With smart detection you can auto-attest CIs that are automatically detected by discovery programs, based only on recent discovery results.

The following conditions must be met to enable smart detection:

- [Discovery](#) is enabled in your organization or [Getting started with Service Graph Connectors](#) are implemented.
- The system property `sn_cmdb_ws.attestation.smart_detection.disabled` is set to **false** (default value).

In addition, smart detection uses the following system properties as filters when creating a list of CIs that are candidates for auto-attestation. To be included as candidates for auto-attestation, CIs must be discovered:

- Within the discovery time window specified by the `sn_cmdb_ws.attestation.smart_detection.discovery_window` system property (for example, within the last 30 days)
- By any discovery source that isn't excluded by the `sn_cmdb_ws.attestation.smart_detection.discovery_source.exclusion` system property

Then, when you [review those candidate CIs](#), you can choose to auto-attest them.

Properties associated with Attestation

Property	Description
<code>sn_cmdb_ws.attestation.smart_detection.disabled</code>	Disables smart detection. • Type: true false • Default value: false • Values: • true: Disable smart detection. • false: Enable smart detection. • Location: Add to System Properties [sys_properties] table. • Learn more: Review CMDB Data Manager attestation tasks in CMDB Workspace
<code>sn_cmdb_ws.attestation.smart_detection.discovery_source.exclusion</code>	Comma-separated list of discovery sources that are excluded in smart detection processing. For example, a data source that is unreliable in detecting CIs.

Property	Description
	CI discovered by discovery sources in the list, can't be candidates for auto-attestation. - Type: string - Default value: Manual Entry - Location: System property - Learn more: Review CMDB Data Manager attestation tasks in CMDB Workspace
sn_cmdb_ws.attestation.smart_detection.discovery_window	Number of days (discovery window) that smart detection uses to determine whether a CI is a candidate for auto-attestation. Only CIs that were discovered within this discovery window can be candidates for auto-attestation. - Type: integer - Default value: 30 - Location: System property - Learn more: Review CMDB Data Manager attestation tasks in CMDB Workspace

Review CMDB Data Manager attestation tasks in CMDB Workspace

Review attestation tasks that are assigned to you or to an assignment group that you belong to in accordance with CMDB Data Manager

Attestation policies. Check the physical existence of IT infrastructure or applications associated with CIs in the attestation task and then process those CIs as appropriate.

About this task

For managers, tasks of direct reports are included, and for user group managers, tasks of members in the group are also included.

Before you begin

Role required:

- CMDB administrator: data_manager_admin or sn_cmdb_admin
- CI and service owner or manager: data_manager_user, sn_cmdb_user (already includes the data_manager_user role)

Canceling an attestation task is available only for tasks that are in the 'In progress' or 'Not started' status, and the required role is data_manager_admin or sn_cmdb_admin.

About this task

After you receive notifications about attestation tasks assigned to you, check the actual IT infrastructure or applications that you own, and then process the CIs in the task accordingly. Initially, all the CIs in the attestation task are listed in a Not Yet Reviewed list and therefore require a review. Attested CIs are then moved to an Attested CIs list, while CIs you reject are moved to a separate Rejected CIs list. After processing all the CIs in the task, you can submit and complete the task. For general information about attestation and the flow of attestation, see [CIs attestation](#).

As you process the CIs in the task, only the fields Attestation Status (values: **Attested**, **Rejected**, **Not Yet Reviewed**), Attested By, and Attested Date are automatically updated. Also, details of your activities are captured in the activity stream of the task.

Attesting tasks is performed in the view in the [My Work view in CMDB Workspace](#).

Procedure

1. Navigate to **Workspaces > CMDB Workspace** and then select **My Work** in the CMDB Workspace menu bar.
Any attestation tasks that are assigned to you or to assignment groups you belong to, appear in the **Attestation tasks** tab in the My tasks section. Attestation tasks are associated with details such as the due dates and short descriptions for those tasks. Assignment groups that you are a member of, are also listed.
2. Select a task to review from the Attestation tasks list.
For a bulk review, select multiple tasks or check the box next to 'Configuration Item' to review all the tasks in the list.
3. Review details of a task:
 - a. (Optional) Administrators can select **Cancel task** if attesting the CIs in the task isn't needed.
For example, if the task was created by error. The canceled task is deleted without any further processing.
 - b. (Optional) If the task isn't properly assigned, select **Reassign** and then select one of the following options.

Depending on your selection, the task might no longer be assigned to you.

- **To me:** If the task isn't yet assigned to a specific user and you are the owner of the CIs who can process the task.

-

To other user: If you aren't the actual owner of the CIs and there is someone else that is more appropriate as the owner for the task. In the Reassign to other user dialog box, set **Assign to** to that user and enter the reason for the reassignment. Then select **Proceed**.

For a CI owner, only users from the assignment group that the task is currently assigned to, appear. For Admins, all users with the data_manager_user role appear.

- **Send for reassignment:** If you aren't the appropriate user for this task and the administrator needs to assign the task to someone else. In the Send for reassignment dialog box, enter

the reason for the operation and then select **Submit** to send a notification to the administrator.

- c. Select **Review CIs to attest** to continue the CI attestation process. If the task is not assigned to anyone, it is automatically assigned to the current user.
4. (Optional) If the Smart detection dialog box appears, select **Review** to review the CIs or **Auto-attest** to attest the detected CIs without a review.
 - a. Review the details of the CIs in the list.
 - b. (Optional) Select **Auto-attest** to automatically attest all the CIs in the list and to move them to the Attested CIs list.
 - c. (Optional) Select **Cancel auto-attestation** to leave all the CIs in the Not Yet Reviewed list and to later continue attesting those CIs regardless of smart detection.

If you select **Cancel**, you can later select **Run auto-attestation** in the Smart detection widget. When reviewing the CIs to attest, the list in the Smart detection pane is filtered to CIs that are candidates for auto-attestation, letting you auto-attest all those CIs in bulk.

5. Review the CIs in the Not Yet Reviewed list.

You can select a CI to open a dashboard with more details such as [CI health](#), key properties, and [CMDB 360 data](#). The CI health details that appear depend on settings and activation of various features.

Use the following cards on the right hand bar to get help and status:



- Select Attestation actions () to get help on the possible actions you can take while reviewing CIs.



- Select Attestation status () to see your progress in reviewing and attesting CIs.



- Select Smart detection status () to see if any CIs are candidates for auto-attestation and to apply auto-attestation if applicable.

6. In the **Details** tab of the Attestation Review CIs pane, select the CIs that you are ready to process and select an action to apply to all selected CIs.

-

Select the CIs that you can attest their existence and select **Attest** to move them to the Attested list.

To undo attestation, select the Attested list, select the CIs that you want to move back to the Not Yet Reviewed list and select **Unattest**. Or, select the CIs that you want to move to the Rejected list and select **Reject**.

-

Select the CIs that you can't attest their existence and select **Reject** to move them to the Rejected list. In the Confirm rejecting CIs dialog box, enter an explanation and then select **Proceed**.

To undo rejection, select the Rejected list, select the CIs that you want to move back to the Not Yet Reviewed list and select **Unattest**. Or, select the CIs that you want to move to the Attested list and select **Attest**.

-

Select the CIs that shouldn't be included in this task and in future Data Manager attestation policies, and select **Exclude**. In the Confirm excluding CIs dialog box, enter an explanation and then select **Proceed**.

Those excluded CIs are added to the exclusion list of CMDB Data Manager attestation policies. For more information, see . Only an admin can undo this operation using the CMDB Data Manager.

-

Select any CIs that you identify as duplicates of other CIs and select **Remove duplicates from task**. In the Confirm removing duplicate CIs dialog box, enter an explanation and then select **Proceed**.

Those duplicate CIs are removed from the attestation task and a de-duplication task is generated for them, which is similar to the de-duplication tasks created by the Identification and Reconciliation Engine (IRE). For more details about how to process and remediate de-duplication tasks, see [Duplicate CIs remediation](#).

Note: Only an admin can undo this operation.

- After processing all the CIs in the task, when the Not Yet Reviewed list is empty, the **Submit** button is enabled. Select **Submit**, and then in the Submit attestation dialog box select **Submit** to close the task.

Data Certification

Data Certification manages scheduled and on-demand validations of data in CMDB and non-CMDB tables.

Information is added to the CMDB by Discovery, by importing from third-party tools, or manually. For regulatory or procedural reasons, information in the CMDB requires checks for accuracy and certification. The person or team responsible for certification can define what information requires verification and the frequency of verification. Using CMDB Data Manager supports policy-driven validation of specific attributes using the certification policy type. CMDB Data Manager generates certification tasks for verifying the data on a recurring schedule, which then, individuals assigned to those tasks answer a series of questions to verify the data.

Use Data Certification to:

- Maintain the accuracy, completeness, and reliability of critical data in both, CMDB and non-CMDB tables.
- Create a framework for ongoing data compliance, regulatory, governance, and operational standards.

Data certification can be performed against specific fields on specific tables. Based on the certification schedule, certification tasks are automatically created and assigned. A certification task represents the work of verifying the data associated with a particular record. For example, you can set up a certification to validate key information fields,

such as **Operating System** and **CPU count**, on all Windows servers located in Chicago. You can then assign the tasks to the appropriate team member automatically.

When planning an implementation of Data Certification, answer the following questions:

- What information requires certification?
- When is the due date for certification?
- Who must perform the certification?

Domain separated systems can use the Data Certification application.

Data Certification experience in CMDB Workspace

You can use the [CMDB Workspace](#) landing page and its views to fully administer and use Data Certification, view various analytics, and handle Data Certification functions such as:

- [Create a certification policy.](#)
- [Convert legacy certification schedules into Data Manager certification policies.](#)
- [Review certification tasks.](#)
- [Review failed certification tasks.](#)

The Data Certification experience in CMDB Workspace doesn't provide functionality such as reset, merging of multiple certification tasks, and escalation.

For more information about using Data Certification in CMDB Workspace, see [Data Certification experience in CMDB Workspace](#).

Data Certification on Core UI (UI 16)

[Data Certification experience in CMDB Workspace](#) provides the latest Data Certification functionality. However, the legacy build of Data Certification on Core UI (UI 16) is available by navigating to **All > Data Certification**. For information about using the CMDB Data Manager legacy build on Core UI, see [Data Certification on Core UI](#).

Note:

Data Certification in CMDB Workspace and the legacy build of Data Certification on Core UI operate separately, and aren't synchronized. Therefore, any Data Certification elements, such as definitions and schedules, that you create in one of those implementations, doesn't appear and isn't included in the other implementation.

To convert data created in the legacy build of Data Certification on Core UI, into draft Certification policies in CMDB Workspace, see [Convert legacy certification schedules into Data Manager certification policies](#).

Data Certification experience in CMDB Workspace

Use Data Certification in CMDB Workspace to administer Data Certification, create certification policies, manage certification tasks, and view analytics about on-going processes.

Note: Data Certification in CMDB Workspace and the legacy build of Data Certification on Core UI operate separately, and aren't synchronized. Therefore, any Data Certification definitions, such as filters and certification schedules, that you create in one of those implementations, doesn't appear and isn't available in the other implementation.

Use Data Certification in CMDB Workspace

Use the [CMDB Workspace](#) landing page and its views to administer Data Certification:

- Use the [Management view in CMDB Workspace](#) to navigate to CMDB Data Manager to [create a new Certification policy](#).

-

Use the My work tile on the CMDB Workspace landing page to access your open and overdue certification tasks that you need to review and process.

-

Use the [My Work view in CMDB Workspace](#) to access any certification tasks assigned to you under the Certification tasks section.

Manage certification tasks in CMDB Workspace:

- [Manage Data Certification instances and tasks, and see progress, status, and insights in the Data Certification Dashboard](#)
- [Review certification tasks in CMDB Workspace](#)
- [Review failed certification tasks in CMDB Workspace](#)

Retention of Data Certification records

Archival and destroy processes of certification policy related records, are separate from those processes for records of all other policy types. This separation facilitates the extension of the retention period of certification policy records. Retaining certification policy execution records for longer periods of time is useful in situations where those records are needed for audits and is also useful for the Data Certification Dashboard, which is populated by these records.

The following configurations facilitate the separation of retention policies for Data Certification records:

- The table cleanup rule for the CMDB Data Management Policy Executions [cmdb_data_management_policy_execution] table, which is stored in the Auto Flushes [sys_auto_flush] table, excludes certification policy execution records from recurring cleanups.
- The Archive CMDB Data Management Tasks archive rule, that applies to all CMDB Data Manager policy execution records, excludes certification policy records. At each archive run, this archive rule is configured to also automatically archive its related records in the CMDB Data Management Certification Task To Document [sn_cmdb_ws_dm_certification_task_to_document] table (Archive Related Records).
- The archive rule, Archive Certification Instances, archives certification policy execution records from the CMDB Data Management Policy Execution [cmdb_data_management_policy_execution] table. This archive rule is configured to archive certification policy execution records 2 years after creation, and to destroy those records 7 years after they are archived.

- The archive rule, Archive Certification tasks, archives the certification task records from the CMDB Data Management Task [cmdb_data_management_task] table. This archive rule also contains an Archive Related Records entry that archives the related records in the CMDB Data Management Certification Task To Document [sn_cmdb_ws_dm_certification_task_to_document] table.

Convert legacy certification schedules into Data Manager certification policies

Convert certification schedules created in the legacy Data Certification application (running on Core UI), into draft Data Manager certification policies available in CMDB Workspace.

Before you begin

Role required: data_manager_admin or sn_cmdb_admin

About this task

If you have been using the legacy Data Certification application on Core UI, then any associated certification schedules aren't automatically available in the new implementation of Data Certification in CMDB Workspace version 6.0. You can convert those definitions into draft Data Manager Certification policies. Then you can publish the converted policies in CMDB Workspace just like publishing any draft Data Manager policy.

Important things to know:

- Legacy certification schedules that were successfully converted are automatically deactivated.
- This conversion migrates most fields from the legacy certification schedule. However, the Assign to empty and the Last run date fields aren't migrated into CMDB Workspace Data Manager tables.
- Any dot-walking settings in the legacy certification schedules used for assignments of User Field or User Group Field are preserved in this conversion.

Procedure

1. Navigate to **All > Workspaces > CMDB Workspace** and then select Management in the CMDB Workspace menu bar.
2. Select the **Data Manager** link in Management tools, in the Manage section.
3. On the Data Manager overview page, select **Import** on the banner at the top of the page.
If there aren't any legacy certification schedules to import, the notification banner about importing existing policies doesn't appear.
4. In the Confirm import of certification schedule into draft policies, select **Import into draft policies**.
 - a. Track progression in the Import in progress dialog box.
 - b. Review the import results in the Import summary dialog box.

If there were earlier conversion operations, the results reflect only the current conversion operation. For example, the Completed count includes only those records that were successfully converted in the current run.

Select the links associated with the completed and skipped results to drill down to the associated records.

Also, to see additional details, select **All** and in the Filter navigator, enter `cert_schedule.list` to open the Certification Schedules table. The Migration Status and the Migration exception columns provide details that are especially important for failed and skipped records.

- c. Select **View draft policies** to view the converted policies in the Data Manager policies page, under the Draft policies tab.
5. (Optional) [Publish a draft CMDB Data Manager policy](#)

Result

- If you log in while this conversion is in process, a progress message appears on the Data Manager overview page. You can select the

View details link to open the Import in progress dialog box with details about the progress of the operation.

- The legacy certification schedules are available as draft Certification policies in Data Manager in CMDB Workspace.
- The legacy certification schedules are deactivated.
- The legacy certification tasks are unchanged and you can review and resolve them as needed.

Data Certification Dashboard

Use the Data Certification Dashboard in CMDB Workspace as a central location to manage Data Certification instances and tasks, and to see counts, progress, state, and analytical insights. You can drill down past due certification tasks to examine and to reassign them, for example, so that they can be completed by an appropriate user.

The screenshot displays the Data Certification Dashboard interface. At the top, there's a navigation bar with 'Home', 'My Work', 'CMDB 360', 'Insights', and 'Management'. Below this, the dashboard title 'Data Certification Dashboard' is visible. The main content area is divided into two tabs: 'Certification instances' (selected) and 'Workload and Aging tasks'. The 'Certification instances' tab shows a list of instances for 'BT Testing 1'. Each instance card includes a 'Policy execution number' (PLCEXEC0000001002), a 'Due date' (e.g., 2025-07-03 10:04:46), and a progress bar indicating '0% complete'. A 'Filter options' section is located at the top left of the dashboard, with dropdowns for 'State' (Not started, In progress), 'Assignment group' (Select), and 'Assigned to' (Select). A 'Search by policy name' field is also present. A 'Certification instance' link is highlighted. The dashboard is titled 'Data Certification Dashboard' and shows 'Showing 1-10 of 18' records.

The Data Certification Dashboard gets populated by Data Certification execution records. For information about the archive rules that control the retention of those records, see 'Retention of Data Certification records' in [Data Certification experience in CMDB Workspace](#).

Role requirements

The user role determines which certification tasks are included in the dashboard, as follows:

- `data_manager_user`: Minimum role required for access. Provides access to those certification tasks that are assigned to the user, to an assignment group that the user is a member of, or if the user is the manager of the assigned user or of the assignment group.
- `data_manager_admin`: Provides access to all certification tasks.

Reviewing certification tasks

The Data Certification Dashboard provides several ways for you to drill down to the certification tasks list, which can contain tasks requiring review. For information about reviewing (and then certifying or failing), reassigning, or canceling tasks, see [Review certification tasks in CMDB Workspace](#).

Certification instances

The Certification instances tab shows counts and other details for certification instances. A certification instance is the collection of certification tasks for a policy execution identified by its unique policy execution number. For each certification instance, the dashboard shows progress indicators such as the overall percent complete, and a general state such as Not Started and Past Due. Each card shows the total number of associated tasks, and within that total, the number of tasks in a complete, open, and unassigned state.

You can:

- Switch the type of view by selecting the List or the Cards icon.
- Filter the list of certification instances that appear, by selecting the available filters such as State, Assignment group, or Assigned to filter and then selecting the specific items that you want to filter by.
- Filter the list of certification instances that appear by searching for specific strings in certification policy names.
- Sort the list of certification instances by various attributes such as instance state, percent complete, or policy execution number. Select ascending or descending sort order to apply along with any list filter.

- Select the info icon next to a certification instance name, to show its respective policy execution number. For information about policy execution numbers, see table CMDB Data Management Policy Executions [cmdb_data_management_policy_execution] in [Components related to CMDB Data Manager](#).

Select a card to see more details about the certification instance, and all the tasks in that instance based on your user role. In the Certification tasks tab you can drill down the certification tasks list to access the tasks themselves for processing.

Select the Certification records tab to see all the certification records associated with all of the certification tasks in the certification instance. You can filter the certification records by **Certified** or **Failed** to show those records with at least one attribute in the selected state.

Workload and Aging tasks

The Workload and Aging tasks tab contains two tabs showing the distribution of certification tasks workload and past-due tasks.

Workload

Shows the distribution of work related to certification tasks, by user groups and by individual users. Select the Groups or the Individuals tab to see the distribution by percentage and by total number of tasks. Select any entry in a list to drill down to the certification tasks list which lets you then access the tasks themselves for processing.

Aging

Shows bar charts with counts of data certification tasks that are past due. Past due state is determined by the setting of the `sn_cmdb_ws.glide.cmdb.data.manager.aging.policy.task.timeframe` system property (by default, 61–90 and over 90 days old). Select a chart to drill down to the certification tasks list which lets you then access the tasks themselves for processing.

Review certification tasks in CMDB Workspace

Review the values of record fields and CI attributes in certification tasks assigned to you, and then certify, edit, or fail the review according to your findings. For managers, tasks of direct reports are included, and for user group managers, tasks of members in the group are also included.

Before you begin

Role required: sn_cmdb_user, sn_cmdb_editor, or sn_cmdb_admin

Canceling a certification task is available only for tasks that are in the 'In progress' or 'Not started' status, and the required role is data_manager_admin or sn_cmdb_admin.

About this task

If a certification task isn't properly assigned, you can reassign it to another user if you know who the appropriate reviewer is, or request an administrator to reassign the task.

Note:

- If the policy is configured to disallow empty attribute values, you will only be able to fail certification of the associated CIs, but you won't be able to certify them unless you populate the empty attributes first.
- If the policy is configured to disallow updates to field values, you won't be able to update non-compliant CIs, resulting in failing their certification.

Procedure

1. Navigate to the Certification tasks list in either of the following ways.
 - Navigate to **Workspaces > CMDB Workspace**, then select **My Work** in the CMDB Workspace menu bar. In the My tasks section, select the Certification tasks tab.
 - Navigate to **Workspaces > CMDB Workspace**. On the Home view in CMDB Workspace, select the **Data Certification Dashboard** quick link. Then, on the Data Certification Dashboard, drill down to the Certification tasks list from a certification instance card or from the Workload and Aging tasks tiles, as described in [Data Certification Dashboard](#).
2. (Optional) In the Certification tasks list, select multiple tasks to reassign tasks in bulk.

- a. Select the Certification tasks tab, then the task or tasks that you want to reassign.
 - Select **Reassign**: Reassign a task to another group or user if you aren't the appropriate user to review the field values and you know who the appropriate reviewer is.
 - **Assignment group**: Select the assignment group the user belongs to.
 - **Assign to**: Search for the appropriate reviewer to reassign this task to.
 - **Submit a reason for this action**: Enter a reason why you are reassigning the task.
 - Select the **Reassign** drop-down menu then select **Assign to me**: Assign the task that isn't assigned, to yourself, if you are the appropriate reviewer for this task.
 - Select the **Reassign** drop-down menu then select **Request reassignment**: Request to reassign the task, that is assigned to you incorrectly, to someone else. Enter a reason why the task needs to be reassigned, then select **Submit**.
- b. Select **Proceed**.
3. In the Certification tasks tab, select the certification task that you want to review.
4. Review the details of the certification task, then select **Review records to certify**.

Note: You can reassign the tasks on this page, see Step 2 to review the reassignment options.
5. Select the records that you want to conclude the review for in the Review not completed tab, and then select **Certify** or **Fail**.

You can select all the records showing on the page for bulk processing by selecting the check box in the first column header, and then selecting the **Select all <n> item/s in this list** link that appears.

- The review conclusion of certified or failed is applied in bulk only to those certification fields in the selected records, which are still pending a review. For example, a certification field that is already marked as certified, isn't affected by selecting **Fail**. Only the unreviewed fields are marked as failed. Records whose review is concluded are moved to the Review completed tab.
 - If the policy is configured to disallow empty attribute values, then CIs with empty fields won't be certified when you select **Certify**. In this situation, to certify those CIs, you must first populate the empty attributes.
6. Select the records that you want to review in details and then select **Review records**.

A check mark next to a column header, denotes the fields whose values are included in this certification review (certification fields). All other columns show additional fields that are included only to help you identify records (display fields).

When you select records (or CIs) to review, the fields (or attributes) for the selected records appear on the right, in a separate field review section. When you select a single record to review, the field values of the selected record appear. When you select multiple records to review, the field values don't appear. In that section, select the fields to apply your review results, or select **Select all fields** to apply the review results to all fields.

Option	Description
Fail selected fields	Select this to denote field values that are incorrect. When you're finished, the records will move to the Review completed tab.
Certify selected fields	Select this to confirm field values that are correct. When you're finished, the records will move to the Review completed tab.
Edit fields	Select this to update the fields whose value is incorrect. When you're finished, select Update .

Option	Description
	You can continue and certify the updated fields.

7. Select **Submit** when all the records have been reviewed.
The **Submit** button isn't available until all the records in the task are reviewed and the task is complete.

What to do next

You can review the certification results in the Review completed tab, and change certification results or edit field values.

Review failed certification tasks in CMDB Workspace

Review records (and CIs) that failed the certification review and are assigned to you. Those exceptions appear in a list that is dynamically updated as certification tasks are being reviewed.

Before you begin

Role required: sn_cmdb_admin

Procedure

1. Navigate to **Workspaces > CMDB Workspace**, then select **My Work** in the CMDB Workspace menu bar.
2. In the Certification exceptions pop-up banner, select **Click here to view**.
3. Select a drop-down to expand a task to show the records and fields that failed the certification review.
4. Select the Certification Document Id of the field that you want to update.
The Element column shows the fields that failed the review. The Certification Document Id column contains links to the respective records.
5. Review the details of fields that require updating.

What to do next

See the [Review certification tasks in CMDB Workspace](#) topic to locate and review the respective certification task and to update incorrect field values. You can also directly access record and CI forms to update incorrect values.

Data Certification on Core UI

The legacy build of Data Certification on Core UI lets you centrally track and handle Data Certification tasks, create certification filters, and define certification schedules.

Starting with the Zurich release, Data Certification plugin (com.snc.certification_v2) is being prepared for future deprecation. It will be hidden and no longer activated on new instances but will continue to be supported. The [CMDB Workspace store app](#) provides the latest experience for this functionality.

- For details about the deprecation, see the [Deprecation Process \[KB0867184\]](#) article in the Now Support knowledge base.
- For information about using Data Certification, see [Data Certification experience in CMDB Workspace](#).
- For information about converting schedules created in the legacy build of Data Certification on Core UI, into draft Certification policies in CMDB Workspace, see [Convert legacy certification schedules into Data Manager certification policies](#).

Note:

Data Certification in CMDB Workspace and the legacy build of Data Certification on Core UI operate separately, and aren't synchronized. Therefore, any Data Certification elements, such as filters and schedules, that you create in one of those implementations, doesn't appear and isn't included in the other implementation.

Activate Data Certification

Activate the Data Certification plugin to access the application. Activating this plugin also activates the Version Management plugin, which manages certification filter versions.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > System Applications > All Available Applications > All**.
2. Find the Data Certification plugin using the filter criteria and search bar.

You can search for the plugin by its name or ID. If you cannot find a plugin, you might have to request it from ServiceNow personnel.

3. Select **Install** to start the installation process.

Note: When domain separation and delegated Admin are enabled in an instance, the administrative user must be in the **global** domain. Otherwise, the following error appears: Application installation is unavailable because another operation is running: Plugin Activation for <plugin name>.

You will see a message after installation is completed. For information about the components installed with a plugin, see [Find components installed with an application](#).

Activating the Data Certification plugin installs the following components.

Demo data is available with Data Certification. The demo data provides information including filters, schedules, instances, and tasks.

Tables

Data Certification adds the following tables:

Table	Description
Certification Audit Definition [cert_audit_definition]	Stores collections of certification schedules that can be run as a single entity.
Certification Audit Definition Elements [m2m_cert_audit_def_cert_sched]	Lists the certification schedules in each certification audit definition.
Certification Audit Instance [cert_audit_instance]	Stores the certification instances associated with a specific audit definition.
Certification Element [cert_element]	Stores the data elements that are grouped into certification tasks.
Certification Filter [cert_filter]	Stores the data that requires certification using a filtering condition for the certification.
Certification Instance [cert_instance]	Stores a collection of certification tasks representing a single instance of a scheduled certification. This table extends the Audit [cert_audit] table.
Certification Schedule [cert_schedule]	Stores certification for a specific set of information on a specific table, what user or group the tasks are assigned to, and how often this certification is done.
Certification Task [cert_task]	Stores individual certification tasks. Certification Task extends the Task table.

Script Includes

Data Certification adds the following script includes:

Name	Description
CertificationAjax	Provides utilities that enable individual certification elements to be certified, rejected, or reverted.
CertificationTaskCreate	Custom code that extends the standard code for certification tasks.
CertTaskEscalationTimerPercentage	Updates time and percentage complete information for a certification.
CertificationUtilities	Provides utility functions for certification.

Client Scripts

Data Certification adds the following client scripts:

Name	Table	Description
Alert If Boxes Checked	Certification Task [cert_task]	Provides a warning if the certifier attempts to leave a record without certifying the checked elements
Check Table Name	Certification Schedule [cert_schedule]	Updates the table name when a different filter is selected.

UI Policies

Data Certification adds the following UI policies:

Name	Table	Description
Hide next scheduled run	Certification Schedule [cert_schedule]	Hides the Next Scheduled Run field when the schedule is set to run once or on demand only.
Hide Run When Not Active	Certification Schedule [cert_schedule]	Hide "run" associated fields when active is set to false.
Make table name read only	Certification Schedule [cert_schedule]	Makes the Table field read-only.
Hide Table field	Certification Element [cert_element]	Hides the Table field on the certification task form.
Make percent complete field read only	Certification Instance [cert_instance]	Makes the Percent complete field read only when the State is Work in Progress, Closed Complete, Closed Incomplete, or Cancelled.
Show Assign to fields	Certification Schedule [cert_schedule]	Shows the Assign To field when the assignment type is User and hides the Assign To field for all other assignment types.
Show Group field	Certification Schedule [cert_schedule]	Shows the Change Group (formerly Assignment Group) field when the assignment type is Group and hides the Change Group

Name	Table	Description
		field for all other assignment types.
Show User field	Certification Schedule [cert_schedule]	Shows the User field when the assignment type is User.
Show Assignment Fields	Certification Schedule [cert_schedule]	Shows the Assign To Empty option when the assignment type is User Field or Group Field.

Business Rules

Data Certification adds the following business rules:

Name	Table	Description
Adjust dates for cert tasks	Certification Instance [cert_instance]	Adjusts dates for tasks belonging to the certification instance when the dates are changed for an active certification.
Cancel Instance	Certification Instance [cert_instance]	Cancels all open certification tasks when an active certification is canceled.
certification audit instance events	Certification Audit Instance [cert_audit_instance]	Sends an inserted event when an active certification audit instance is created. Sends a completed event when an active certification audit instance is

Name	Table	Description
		marked as complete or incomplete.
certification element events	Certification Element [cert_element]	Sends a failed event when an element of a certification is marked as failed.
certification instance events	Certification Instance [cert_instance]	Sends an inserted event when an instance of a certification is created. Sends a completed event when an instance of a certification is completed.
Certification Instance Rollup	Certification Task [cert_task]	Updates the Percent complete field on the certification instance record.
certification task events	Certification Task [cert_task]	Sends an inserted event when a task is inserted. Sends a completed event when a task is deactivated. Sends a canceled event when a task is canceled.
Certification Task Values	Certification Element [cert_element]	Updates the percent complete of the parent task when a certification element is updated.
Check Certification Audit Progress	Certification Instance [cert_instance]	Updates the completion status of the audit instance

Name	Table	Description
		as a whole when a certification that is part of an audit is complete.
Clean Certification Views	Certification Instance [cert_instance]	Cleans all related records when a certification instance is deleted.
Copy certification schedule fields	Certification Instance [cert_instance]	Copies changes to the certification schedule to the certification instance.
Merge Certification Tasks	Certification Task [cert_task]	Merges two tasks together when a task is reassigned and there is another task for the same instance with the new user.
Prevent delete of Filter with Schedule	Certification Filter [cert_filter]	Prevents the deletion of a filter that is used in a schedule.
Reassign Notification	Certification Task [cert_task]	Sends out a notification to the new and previous assignees when a task is reassigned.
Rollup State	Certification Task [cert_task]	Updates all necessary parent items when task state is changed.
Update audit reference	Certification Task [cert_task]	Makes Data Certification records compatible with Desired State records. This rule makes sure

Name	Table	Description
		that the Audit field is correctly completed when a record is inserted using Insert and Stay.
Update audit result	Certification Element [cert_element]	Makes Data Certification records compatible with Desired State records for reporting purposes. This rule puts certified values in the Desired value column when an audit is Certified. It also puts actual values in the Discrepancy value column when an audit is Failed.
Update follow_on_task & audit references	Certification Element [cert_element]	Makes Data Certification records compatible with Desired State records for reporting purposes. This rule makes certification tasks compatible with follow-on tasks and displays all tasks, regardless of origin.
Update next run time	Certification Schedule [cert_schedule]	Updates the Next scheduled run field when a schedule runs Daily, Weekly, Monthly, or Periodically.
Verify Fields	Certification Schedule [cert_schedule]	Verifies that no field is used in both Display and Certification fields

Name	Table	Description
		when the fields of a certification schedule are changed.

Formatter

Data Certification adds the following formatter:

Formatter

Name	Description
Certification Task Elements	Enables custom user interface formatting of elements on a certification task. For example, displays the green check mark and red exclamation point to use when certifying an element.

Properties

Properties

Name	Table	Description
glide.ui.cert_task_activity.fields	System Properties [sys_properties]	Defines which journal field is the task activity field. Default: work_notes

User Roles

Data Certification adds the following user roles:

User Roles

Role	Contains Roles	Description
certification_admin	certification	Can:

Role	Contains Roles	Description
		• Create and configure certifications • Override provided answers • Perform certification tasks for certification task owners • Send certification task notifications to users and owners at any time • Cancel or delete certifications in any state
certification_filter_admin	certification	Can create and manage all data certification filters.
certification	none	Can update active or incomplete tasks assigned to them or to groups of which they are a member. Can also update configuration items owned by them or by groups of which they are a member. Receives email notifications when assigned certification tasks.

Events

Data Certification adds the following events. The ServiceNow system uses these events to send email notifications to task owners and managers about changes in certification records.

Name	Description
cert_audit_instance.completed	A certification audit instance has been completed.
cert_audit_instance.inserted	A certification audit instance has been inserted.
cert_element.failed	A certification element has failed certification.
cert_instance.complete	A certification instance has been completed.
cert_instance.inserted	A certification instance has been inserted.
cert_task.cancelled	A certification task has been canceled.
cert_task.completed	A certification task has been completed.
cert_task.escalate	A certification task record has been escalated.
cert_task.inserted	A new certification task has been created.
cert_task.notifications	A certification task notification has been resent to a user.
cert_task.overdue	A certification task is past its specified completion date.

Name	Description
cert_task.reassign	A certification task has been reassigned.
cert_task.warning	A new task escalation point has been reached.

Email Templates

Data Certification adds the following email templates:

Name	Message
certification.task.cancelled	A certification task assigned to you/your group as part of the data certification and management process has been canceled.
certification.task.reminder.inserted	A certification task that has been assigned to you/your group as part of the data certification and management process requires attention.
certification.task.reminder.outstanding	A certification task that has been assigned to you/your group as part of the data certification and management process requires attention.
certification.task.reminder.overdue	A certification task that has been assigned to you/your group as part of the data certification and management process is overdue.

Certification schedules

A certification schedule defines the information that requires certification and the frequency of execution.

At each time interval specified, or on-demand, the certification schedule generates a set of certification tasks based on set conditions. Use the Preview Certification Tasks related link to preview the certification tasks generated from a certification schedule.

Certification schedule

The screenshot shows the 'Certification Schedule' configuration page for 'Unix Server in South Africa'. The interface includes a header with navigation icons and buttons for 'Update', 'Execute Now', and 'Delete'. The main configuration area is divided into several sections:

- Name:** Unix Server in South Africa
- Filter:** Unix Server in South Africa - 1
- Table:** UNIX Server [cmdb_ci_unix_s...]
- Active:** ☒
- Run:** Monthly
- Day:** 1
- Time:** Hours 00, 00, 00
- Last run date:** 2012-10-29 12:22:07
- Next scheduled run:** 2012-11-01 00:00:00
- Display fields:** ☐ (locked icon)
- Certification fields:** ☐ CPU speed (MHz), RAM (MB), CPU core count
- Assignment type:** Specific User
- User:** Abel Tuter
- Days to complete:** 2
- Task description:** Ensure that all Unix Servers in South America data center meet minimum reqs
- Instructions:** A rich text editor containing the text: 'The minimum requirements for Unix Servers are: CPU Speed: 3ghz, Ram: 16gb, CPU Core Count: 4'.

At the bottom, there are buttons for 'Update', 'Execute Now', and 'Delete'.

Certification tasks

A certification task represents the work of verifying the data associated with a particular record.

Task owners are responsible for performing the certification tasks. Tasks have an associated workflow that sends reminders to the task owner and, if necessary, the manager of the owner at regular intervals.

Certification task

< ≡ Certification Task
TSK0009072

... Follow Update Delete ↑

Number	TSK0009072	Complete by	2018-11-13 15:04:57
Assigned to	Abel Tuter	State	Closed Complete
Assignment group		Percent complete	100
		Escalation	Normal

Short description

Ensure that all Unix Servers in South America data center meet minimum reqs	center meet minimum reqs
--	---

Work notes

Activities: 0

No activity

Certification data for Certification Task TSK0009072

≡ UNIX Servers New Go to

<< < 1 to 3 of 3 > >>

All

	CPU speed (MHz)	RAM (MB)	CPU core count
☐	1,650	8,192	1
☐	1,650	8,192	1
☐	1,650	8,192	1

Note: If the message

Record cannot be certified until the instance is finished creating all certification tasks and elements. Reload the page to try again

appears, it signifies that:

- A large amount of data is present in the cmbd_ci and cmbd_ci_server tables.
- Data certification task processing is not complete (Data Certification jobs are still in process).

As directed, reload the page and wait for the processing to complete.

Clean up invalid elements

Use the **Clean up invalid elements** UI action to query and delete certification elements that reference invalid records. Each certification task has a certification schedule, and each certification schedule has Table and Filter fields. When you use this UI action, it performs the following processing:

1. Collects all available records from Table field in the certification schedule with filters that are available in certification schedule.
2. Collects all certification elements associated with the current certification task.
3. Deletes the certification elements that are no longer available for the data collected in the previous step.
4. After deleting invalid records, it recomputes the certification completion percentage using the following formula:

$$(1 - (\text{number of certification elements pending} / \text{total no of certification elements associated})) * 100;$$
5. If there are no certification elements with a Pending status, it marks the associated certification task as Closed, and deactivates it.
6. If there are remaining certification elements with a Pending status, it activates the associated certification task and changes its status to Work in Progress.

Certification elements

Each element of each record being certified is tracked in its own certification element record.

Also tracked are the date and time when the element was certified, comments, and the original and certified values of the field. You can view elements on individual certification tasks. Or, you can directly access the Certification Elements [cert_element] table by navigating to **All** and then in the Filter box, entering `cert_element.list`.

Certification elements

Certification Elements

New

Go to

Certified

Search

1 to 7 of 7

All

258 total Certification Elements

Certified

Document

State

Element

Original value

Certified value

Comment

Certification Task

Audit

Search

Search

Search

Search

Search

Search

Search

Search

Search

Audit: Certify groups (8)

Audit: Certify servers (189)

Audit: IBM Server Schedule (7)

Audit: Oracle Database Schedule (5)

☐		2015-08-29 15:10:02	Database: bond_trade_uk	Certified	change_control	TSK0009071	Oracle Database Schedule
☐		2015-08-29 15:10:02	Database: bond_trade_ny	Certified	change_control	TSK0009071	Oracle Database Schedule
☐		2015-08-29 15:10:06	Database: SAP ORA01	Failed	change_control	TSK0009071	Oracle Database Schedule
☐		2015-08-29 15:10:02	Database: NY RAC	Certified	change_control	TSK0009071	Oracle Database Schedule
☐		2015-08-29 15:10:02	Database: PS ORA01	Certified	change_control	TSK0009071	Oracle Database Schedule

Audit: Unix Server in South Africa (9)

Audit: Verify Oracle (5)

Audit: Verify User Lenovo Computers (64)

Certification instances

A certification instance is the collection of certification tasks for one execution of a certification schedule.

Certification instances

Certification Instances

Go to

Number

Search

<

Certification audit instances

A certification audit instance is a collection of the certification instances and tasks generated by a single execution of the certification audit definition.

Certification audit instance

Certification Audit Instances				
Go to		Number	Search	1 to 1 of 1
All	Number	Certification Audit Definition	Created	Complete by
Search	Search	Search	Search	Search
☐	CRTAUD00000001	Certify Servers and Groups	2018-10-30 08:09:36	2018-11-13 07:09:36

Certification audit definition

A certification audit definition is a collection of certification schedules that can be run at once.

Certification audit definition

<

≡

Certification Audit Definition
Certify Servers and Groups

Update

Delete

Name

Certify Servers and Groups

Days to complete

14

Description

Annual certification of Servers and Groups

Update

Delete

Related Links

[Create Certification Audit Instance](#)

Certification Schedules (2)

Certification Audit Instances (1)

≡

Certification Schedules

New

Edit...

Go to

Certification Schedule

▼

Search

◀◀

◀

1

to 2 of 2

▶

▶▶

⌵

Certification Audit Definition = Certify Servers and Groups

≡ Certification Schedule

≡ Table

≡ Certification fields

☐

[Certify groups](#)

[sys_user_group](#)

cost_center,email,type,default_assignee

☐

[Certify servers](#)

[cmdb_ci_server](#)

classification,cost_center,owned_by,support_group

Data Certification Overview module

The Data Certification Overview module displays various data certification-related reports on the Data Certification Console homepage.

The Overview module is a type of homepage.

The different levels of access are:

Access levels per role

Role	Access
certification	View (view overview page and refresh reports)

Role	Access
	Review, certify, and fail certifications
certification_admin	- View (view overview page and refresh reports) - Customize (refresh, add, delete, and rearrange reports) View, customize
admin	- View (view overview page and refresh reports) - Customize (refresh, add, delete, and rearrange reports) - Edit (can edit reports)

Data Certification Overview Module

The Overview module includes the following reports:

Data Certification Overview Module Description

report	Description	Table
30/60/90 Day Aging	Groups tasks by the number of days (30, 60, 90, and 90 and over) since the task was opened.	Certification Task
Certification Instances	Lists all certification instances.	Certification Instance
Certification Progress Report	Groups tasks by task owner, indicating	Certification Task

report	Description	Table
	task progress as a percentage.	
Certification Task Completed Report	Groups tasks by task owner, indicating tasks that are complete.	Certification Task
Exceptions To Date	Lists all task elements that have comments added and a state of Failed or In Progress.	Certification Element
Functional Roll Up	Lists the managers that have groups with assigned certification tasks. The report is a horizontal bar chart, grouped by status, with each bar representing a manager of an assignment group.	Certification Task
Hierarchical Roll Up	Shows the managers that have employees with assigned certification tasks (task owners). The report is a horizontal bar chart, grouped by status, with each bar representing a manager of a task owner (identified in the Assigned to field).	Certification Task
Upcoming Schedules	Lists all schedules that are scheduled to run within the next 30 days.	Certification Schedule

Related tasks

- [Use the Data Certification Overview module](#)

View the status of data certification tasks.

Before you begin

Role required: admin

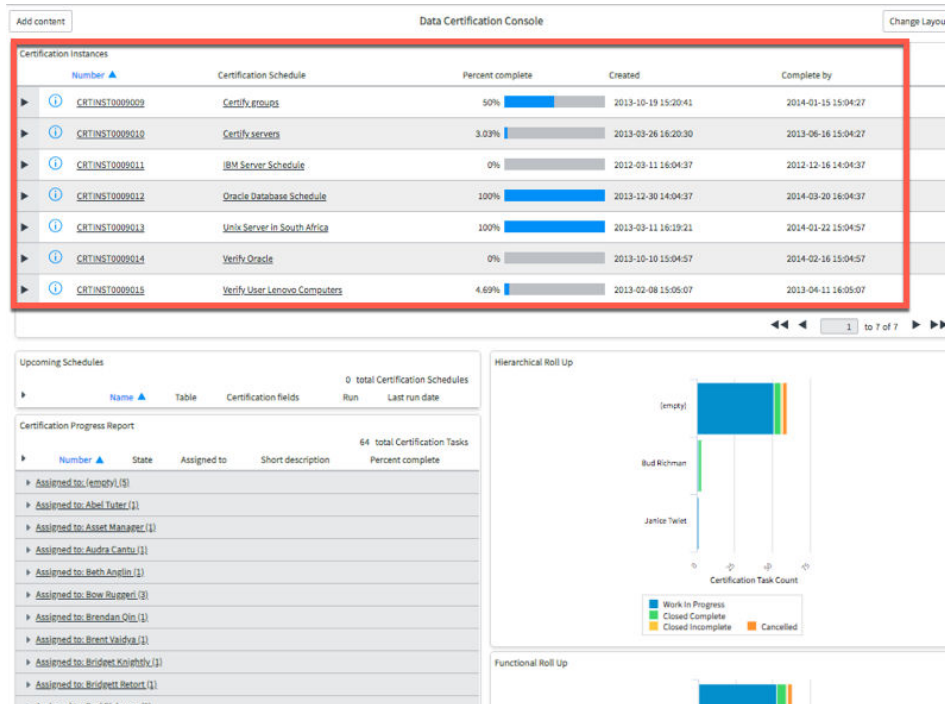
Procedure

1. Navigate to **All > Data Certification > Overview**.
2. Click elements within the reports to obtain more information.
For example, click any of the colored bars in the **Functional Roll Up** bar chart and detailed information replaces the Data Certification Console screen.
3. Update some fields directly on the overview page.

Example

For example, in the red box on the image shown, a certification schedule is being updated in the certification instances report.

Data certification overview module



Related reference

- [Data Certification Overview module](#)

Data Certification planning

Initial planning can make the certification process more successful.

By defining certification schedules and certification audit definitions, users with the certification_admin role establish when certifications are performed, who performs it, and what data must be certified.

Required Roles

Users with the certification_admin role can view filter versions. These users can create, update, and delete filters, if they have the proper access to necessary tables. In the base ServiceNow system, certification_admin users have limited system rights and do not have access to all the tables required for creating a filter. When assigning compliance resources,

make sure to grant additional roles to the certification_admin user as needed. For example, this user requires roles that grant access to these tables:

- Company [core_company]
- Cost Center [cmn_cost_center]
- Schedule [cmn_schedule]

Planning Data Certification

Planning the data certification process requires defining:

- The certification schedule defines certification for a particular set of information on a particular table. It also generates certification tasks to perform that certification. One certification task is generated per task owner and a certification instance record groups the tasks.
- The optional certification audit definition groups some certification schedules to be performed together and generates certification audit instances to perform them.

The following questions require answers for each certification schedule:

- What information requires certification?
- When is the due date for certification?
- Who must perform the certification?

A filter is a subset of configuration items from any ServiceNow table that is created with a standard condition builder.

Before you begin

Role required: admin

About this task

An example is a filter that selects all UNIX servers in the Australian data center.

With filters, you can:

- Create multiple versions of a filter and then select the version you want to use.
- Use one filter on multiple certification schedules.
- View the number of records that match your filter as you create the conditions.

Note: Be sure to create certification filters before creating certification schedules.

Procedure

1. Navigate to **All > Data Certification > Certification Filters**.
2. Click **New**.
3. Fill in the form (see table).
4. Click **Submit**.

This action saves the filter as version 1.

Certification filter V1

The screenshot shows the 'Certification Filter' form in ServiceNow. The form is titled 'Certification Filter New record'. It includes the following fields and controls:

- Number:** CFLR0010001
- Active:** ☒
- Name:** Linux Servers
- Description:** Dell Linux servers
- Table:** Linux Server [cmdb_ci_linux...]
- Version:** 1
- Filter condition:** 4 records match condition
- Buttons:** Add Filter Condition, Add "OR" Clause
- Filter Rule:** Manufacturer is Dell Inc.
- Submit Button:** Submit

5. To create another filter version, modify the filter conditions and click **Update**.
The system saves the new filter and increments the version number.

Certification filter V2

Number: CFLR0010001 Active: ☒

Name: Linux Servers

Description: Dell Linux servers

Table: Linux Server [cmdb_ci_linux_ser...] Version: 2

Filter condition: 0 records match condition

Add Filter Condition Add "OR" Clause

All of these conditions must be met

Manufacturer is Dell Inc. AND OR X

Operating System is Linux Red Hat AND OR X

By default, the Certification Filters list shows only the current version of each filter. To see all filter versions, click **All** in the breadcrumbs.

Certification Filter List

Certification Filters New Go to Name Search 1 to 15 of 15

All Displays all versions

Name	Description	Table	Version
Unix Servers	Only Unix servers	Server [cmdb_ci_server]	1
Unix Server in South Africa	Only Unix Servers in South Africa	UNIX Server [cmdb_ci_unix_server]	1
Oracle Database	Only Oracle Databases	Database [cmdb_ci_database]	1
Linux Servers	Dell Linux servers	Linux Server [cmdb_ci_linux_server]	2
Linux Servers	Dell Linux servers	Linux Server [cmdb_ci_linux_server]	1
IBM Servers	Only IBM Servers	Server [cmdb_ci_server]	1
Data Center Zones	All data center zones	Data Center Zone [cmdb_ci_zone]	1

- To make an inactive filter the current version, open the inactive filter and click **Revert**.

Certification filter revert

Number: CFLR0010001 Active: ☐

Name: Linux Servers

Description: Dell Linux servers

Table: Linux Server [cmdb_ci_linux...] Version: 1

Filter condition: 0 records match condition

Add Filter Condition Add "OR" Clause

Manufacturer is Dell Inc.

Revert Delete

Related Links: [Delete inactive versions](#)

Reverts an inactive version to the current version

This action creates a new, active version of the filter and makes all previous versions inactive.

7. To delete a single filter version, open that version record and click **Delete**.
8. To delete inactive versions of a filter, click **Delete inactive versions** under **Related Links** in that filter record.
You cannot delete a filter that is used in a schedule definition. The system displays a warning and the filter is not deleted.

Creating certification filters

Field	Description
Name	[Required] Filter name.
Description	[Optional] Brief description of the filter.
Number	[Read-only] Automatically assigned filter identification number.
Table	Table containing the records to be filtered. Use of the Database View [sys_db_view] table is limited by version.

Field	Description
Active	Control to make the filter available for use from the Filter field on the Certification Schedule form.
Version	Current version of this filter. Any significant changes to the filter make the current version inactive. The system copies the updated filter, marks it as active, and increments the version number. The system saves all versions of the filter and makes them available to users. More than one version of a filter can be marked active.
Filter condition	Field, operator, and value to create the condition. The available options depend on the table selected. You can view the number of records that match the filter by clicking the refresh icon. Refresh Conditions  If the filter does not match any records, the system marks the certification instance as Closed Complete, with the Percent complete value set to 100%.

Related concepts

- [Data Certification planning](#)

A certification schedule specifies the fields to display, the fields that require certification, certification task assignments, completion requirements for task owners, frequency of schedule, and detailed instructions.

Before you begin

Role required: admin

About this task

Use the preview option to see what tasks are created before saving the schedule. If the tasks are not what you want, edit the schedule and preview the tasks again. The system creates certification tasks automatically when it executes a schedule.

To schedule a certification:

Procedure

1. Navigate to **All > Data Certification > Schedule Definitions**.
2. Click **New**
3. Fill in the fields (see table).
4. Click **Submit**.

Defining A Certification Schedule

Field	Description
Name	A schedule name.
Filter	A certification filter for this schedule.
Table	[Read-only] The table holding the records to be certified. To change the table name, select

Field	Description
	a different Filter or create a new Filter .
Display fields	The fields displayed in the Certification Task list to provide context. These do not require certification themselves. For example, although users are not required to certify the Name field of a record, it displays so that users know what record they are certifying.
Certification fields	The fields to certify on this certification schedule.
Assignment type	A choice list to select how the certification schedule assigns the certification tasks. • User Field: Select a user reference field on the table being certified. As an example, select the user named in the Managed by field to identify the user who performs the task. This selection displays the Assign to and Assign to empty fields. If the reference field on the record is empty, the value in the Assign to empty field is used. • Specific User: Select a specific user to perform the tasks. This selection displays the User field. • Group Field: Select a group reference field on the table being certified. As an

Field	Description
	example, select the Support group field to identify the user who performs the task. This selection displays the Assign to group and Assign to empty fields. All members of the group from the reference field on the record are assigned to the tasks. If the reference field on the record is empty, the value in the Assign to empty field is used. • Specific Group: Select a specific group to perform the tasks. This selection displays the Group field. All members of the named group are assigned to the tasks.
User	This field appears when: • Assignment type is Specific User. This system assigns this user to all certification tasks for this schedule. • The Assign to empty field is set to Create Assigned Task, and you have selected User Field as the assignment type. The system assigns this user to certification tasks containing unassigned records. You can only select users with the certification role.

Field	Description
Assign to group	The group field that defines the group assigned to the certification tasks. This field is available only when the Assignment type is Group Field .
Group	The specific group to which certification tasks are assigned for this schedule. This field is available only when the Assignment type is Specific Group
Assign to	The user field that defines which user is assigned to the certification task. This field is available only when the Assignment type is User Field .
Assign to empty	The behavior to use if the field selected in Assign to or Assign to group is blank on the record being certified. For example, if a task must be assigned to a manager, but no manager is identified, the value in this field determines what happens. This field appears only when the Assignment type is User Field or Group Field. The possible selections are: • Do Not Create Task: No task is created when the Assign to or Assign to group field is empty. • Create Unassigned Task: Create a task, but do not assign it to any user or group.

Field	Description
	The task can be manually assigned later. • Create Assigned Task: Create a task and assign it to the user or group specified. If you selected an assignment type of User Field, the User field is available. If you selected the Group Field type, the Group field is available. The schedule automatically creates certification tasks for all records that do have "Assign to" populated, regardless of which selection you make for "Assign to empty."
Days to complete	[Required] The number of days that task owners have to complete the certification tasks. When the certification schedule is part of a certification audit definition, the Days to Complete audit definition value overrides the value set for the certification schedule.
Active	Check box to activate this certification schedule, generating certification tasks at the scheduled date and time. Clear this check box to hide scheduling fields on the form (except Last run date) and not generate certification tasks.

Field	Description
Run	How often to run the schedule that generates certification tasks: • Daily • Weekly • Monthly • Periodically • Once • On Demand
Day	When Run is Weekly, the day of the week when the schedule runs and generates certification tasks. When Run is Monthly, the day of the month the schedule runs and generates certification tasks. If the day is 29, 30 or 31, the certification runs on the last day of the month for shorter months.
Repeat Interval	When Run is Periodically, the frequency that the schedule runs to generate certification tasks, entered in time, days, or both. For example, set Days to 10 and Hours to 14:00:00 to run the schedule and generate certification tasks every 10 days at 14:00.
Starting	When Run is Periodically or Once, the date and time the

Field	Description
	schedule runs and generates certification tasks.
Time	When Run is Daily , Weekly , Monthly , or Once , the time of day, on a 24-hour clock, the schedule runs and generates certification tasks.
Last run date	[Read-only] The date and time that the schedule ran last, either on its regular schedule or manually, and generated certification tasks.
Next scheduled run	[Read-only] The next date and time the schedule runs and generates certification tasks.
Task Description	A description to add to the Short Description field of the certification task.
Instructions	An HTML field for providing instructions to the user or group performing the certification.

Related concepts

- [Data Certification planning](#)

Previewing certification tasks saves any changes to the Certification Schedule form and displays the tasks that are created when you execute the certification schedule.

Before you begin

Role required: admin

About this task

Previewing tasks is especially useful if you want to test different combinations of options in the **Assignment type**, **Assign to**, and **Assign to empty** fields.

Procedure

1. Navigate to **All > Data Certification > Schedules > Schedule Definitions**.
2. Click a certification schedule **Name**.
3. In **Related Links**, click **Preview Certification Tasks**.
The tasks to be created appear at the top of the screen.
Certification info message

 0 Servers

One Certification Task would be created unassigned to certify 31 Servers
One Certification Task would be created for Bow Ruggeri to certify 7 Servers
One Certification Task would be created for Bud Richman to certify 1 Server
One Certification Task would be created for David Loo to certify 1 Server



Related tasks

- [Define a certification schedule](#)

Related concepts

- [Data Certification planning](#)

After you define a certification schedule, the system automatically sends notifications to specific users based on the information in the schedule.

Before you begin

Role required: admin

About this task

The following notifications are sent automatically:

Certification Schedule Notifications

Time elapsed to end date	Email template name	Notification message is sent to
0% (when task is created)	certification.task.reminder.inserted	Task owner or assignment group, if specified
50%	certification.task.reminder.outstanding	Task owner or assignment group, if specified
75%	certification.task.reminder.outstanding	Task owner, assignment group, if specified, and manager of the task owner, if specified
95%	certification.task.reminder.outstanding	Task owner, assignment group, if specified, and manager of the task owner, if specified
100%	certification.task.reminder.overdue	Task owner, assignment group, if specified, and manager of the task owner, if specified

The email templates used in the notifications can be edited, for example, to change the email message text.

Executing a Certification Schedule

Executing a certification schedule generates certification tasks based on the schedule.

Procedure

1. Navigate to **All > Data Certification > Schedules > Schedule Definitions**.
2. Click a certification schedule **Name**.
3. Click **Execute Now**.
The related lists **Certification Instances** and **Certification Tasks** display the instances or tasks generated by the schedule. The amount of time it takes to generate all certification tasks depends on the size of the table selected and how many fields require certification.
Execute certification schedule

Number	State	Assigned to	Percent complete
TSK0009065	Work In Progress		0%
TSK0009067	Cancelled	Bow Ruggeri	0%
TSK0009068	Work In Progress	Bud Richman	0%
TSK0009069	Closed Complete	David Loo	100%

A certification audit is a collection of certification schedules that can be run as a single entity.

Before you begin

Role required: admin

About this task

Certification audits can be useful when there are multiple certification schedules. After creating a certification audit definition, you can generate a certification audit instance. The certification audit instance is a collection of the certification instances and tasks generated by a single execution of the certification audit definition.

Procedure

1. Navigate to **All > Data Certification > Audits > Audit Definitions**.

2. Click **New**.
3. Fill in the fields (see table).
4. Right-click the header bar and select **Save**.
5. In the **Certification Schedules** related list, click **Edit**.
6. In the **Collection** list on the left, select one or more schedules and click **Add**.
7. Click **Save**.
8. In **Related Links**, click **Create Certification Audit Instance**.
The system generates an audit instance based on the certification schedules selected. All audit instances based on this audit definition are listed in the **Certification Audit Instances** related list.

Defining and Creating a Certification Audit

Field	Description
Name	The name of the audit definition.
Days to Complete	The number of days that task owners have to complete the certification tasks created by this audit definition. Overrides the identical field on the certification schedule.
Description	A short description of the intended audit.

Related concepts

- [Data Certification planning](#)

You can view a list of all certification audit instances at any time.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Data Certification > Audits > Audit Instances**.
2. View the **Certification Instances** related list.

The list contains each of the associated instances generated as part of the audit.

Related concepts

- [Data Certification planning](#)

Data certification performance

After the certification process has been planned, certification tasks can be performed according to defined schedules.

Users with the certification role can perform certification tasks. The certification tasks can be tracked as part of certification instances.

Related tasks

- [Define a certification schedule](#)

Related concepts

- [Data Certification planning](#)

Related reference

- [Installed With Data Certification](#)

After you execute a certification schedule manually or at a scheduled time, the ServiceNow system performs certain actions.

- Creates certification tasks for any records that meet the filter requirements in the specified table, like tasks from the Configuration Item [cmdb_ci] table.

- Assigns the new tasks to the user or group identified in one of these [certification schedule](#) fields:
 - Assign to
 - User
 - Assign to group
 - Group
- Places the new tasks in the Work in Progress state.
- Adds the certification schedule Short description and Assigned to values to the corresponding fields on the certification task record.
- Adds the certification schedule Days to complete and Complete by date fields to the certification task record, based on when the task is created.

Note: If the certification filter does not match any CIs, the system sets the State to Closed Complete and the Percent complete to 100.

To view tasks assigned to you, navigate to **Data Certification > Tasks > My Tasks**. To resolve tasks assigned to you, see [Certify an element](#). For more information about purpose and usage of certification tasks, see [certification tasks](#).

The following information is tracked on the certification task record:

Certification task record

Field	Description
Number	An identification number for the certification task.
Assigned to	The user responsible for certifying the data.
Assignment group	The group responsible for certifying the data.

Field	Description
Complete by	[Read-only] A date field containing a deadline for the task. This field is automatically filled in based on the Days to Complete field on the certification schedule.
State	[Read-only] The current state of the certification task. The selections are: Work in Progress, Closed Incomplete, Closed Complete, and Cancelled.
Percent complete	The task progress as a percentage. This field is read-only when a task is in a Closed Incomplete, Closed Complete, or Cancelled state.
Escalation	[Read-only] The escalation level of the task. When 0–49% of the time to Complete By has elapsed, this field is set to Normal. At 50%, this field changes to Moderate and an email reminder is sent to the task owner. At 75%, this field changes to High and an email reminder are sent to the task owner and the manager of the task owner. At 95%, this field remains set to High, but a second email reminder is sent to the task owner and manager.
Short Description	A short description of the task. This field is automatically filled in with the text from the certification schedule of the Task description field.

Field	Description
Work notes	Information about work performed on the certification.

The Certification Task form contains a list of all elements to be certified.

Before you begin

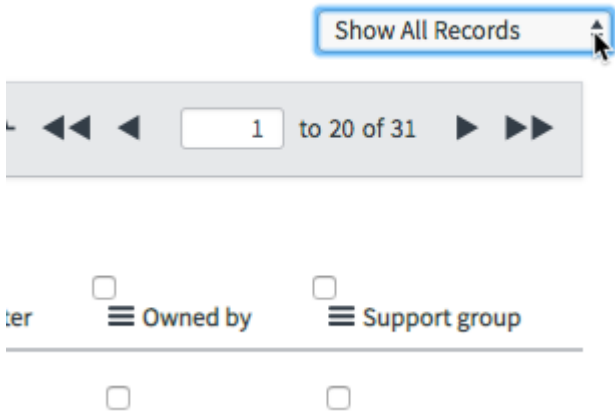
Role required: admin

About this task

Note: After you certify all the elements in a task, no elements can be reverted.

Procedure

- 1. Navigate to **All > Data Certification > Tasks > My Tasks**.
 - 2. Open a certification task with a State of Work in Progress.
 - 3. In the upper right corner of the list, select records that require certification for this task or all records that are part of this certification task.
- Certification list 3



4. Select the check box beside a certification element.
5. In Optional comment for checked elements, above the list, enter information that would be useful to others.
Certification list

Certifications required for Certification Task TSK0009065 [\(toggle help\)](#)

Server not at the designated location  

 Servers [New](#) Go to 

 All


 Name 
 Location

☒	 ApplicationServerPeopleSoft	5052 Clairemont Drive, S
☐	 AS400	4193 University Avenue,

6. Do:
 - Click the green check mark to certify the element.
 - Click the red exclamation point to fail the element.
7. To see the certified or failed element, set the view to Show All Records.

A green check mark or red exclamation mark appears beside the element.

8. Point to an icon to see any certification comments.
9. Ensure that all elements have the correct certification, either accepted or rejected.

After you certify all elements, no elements can be [reverted](#). When all elements of a certification task are certified or rejected, the task State changes to Closed Complete.

View audit results after you certify the elements.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Data Certification > Schedules > Audit Results**.

The list of data certification audit results appears, grouped by certification instances. Certified configuration items show the Original value only. Failed CIs contain the Certified value and the Original value.

2. Click the links in the list to open any of the related records.

Data cert audit results

Audit	Date	Database	Status	Element	Original value	Certified value	Comment	Certification Task	Audit
Audit: Certify groups (8)									
Audit: Certify servers (160)									
Audit: Certify servers (160)									
Audit: IBM Server Schedule (7)									
Audit: Oracle Database Schedule (5)	2015-08-29 15:10:02	Database: bond_trade_uk	Certified	change_control				TSK0009071	Oracle Database Schedule
	2015-08-29 15:10:02	Database: bond_trade_ny	Certified	change_control				TSK0009071	Oracle Database Schedule
	2015-08-29 15:10:06	Database: SAP ORA01	Failed	change_control				TSK0009071	Oracle Database Schedule
	2015-08-29 15:10:02	Database: NY RAC	Certified	change_control				TSK0009071	Oracle Database Schedule
	2015-08-29 15:10:02	Database: PS ORA01	Certified	change_control				TSK0009071	Oracle Database Schedule
Audit: Unix Server in South Africa (8)									
Audit: Verify Oracle (5)									
Audit: Verify User Lenovo Computers (64)									

Users with the certification_admin role can export the certifications list and save the list in Excel, CSV, XML, or PDF format. This list is useful when you have a long list of certification elements or if many different users are assigned to certify elements on a single certification schedule.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Data Certification > Tasks > All Tasks**

2. Open a task.
3. Open any column context menu in the certification data list and complete the export.

You cannot reset any element after all elements are certified.

- To reset individual certifications, right-click the element in the certification list and select **Revert Certification**.
- To reset the entire task to its starting point, click the **Reset all Certifications to Pending** related link.

The Certification Tasks related list on the certification instance record provides information about associated tasks.

Before you begin

Role required: admin

About this task

The State field on the certification instance record is read-only and is based on the cumulative states of the certification tasks associated with the instance. The Percent complete column allows users with the certification_admin role to track task progress quickly. For more information, see [Track Certification Tasks](#).

To track a certification instance:

Procedure

1. Navigate to **All > Data Certification > Schedules > Instances**.
2. Click a certification instance Number.
3. View and edit the following fields as necessary.

Certification instance

Field	Description
Number	[Read-only] Automatically generated identification number for the instance.
Certification Schedule	The certification schedule used to create the certification instance.
State	[Read-only] Current state of the certification instance: Work in Progress, Complete, Closed Incomplete, or Cancelled. For more information, see Track Certification Tasks .
Created	[Read-only] Date and time the certification instance was created. Date is filled in automatically when the Execute Now button clicks the associated certification schedule.
Complete by	[Required] Date and time when the certification instance must be completed. The system updates this field when it executes the schedule, using the deadline specified on the instance. All certification tasks associated with the certification instance must be marked Complete, Closed Incomplete, or Cancelled before the instance is complete.
Percent complete	Percentage of the instance that has reached the Closed Complete state. This field is automatically filled in based

Field	Description
	on the Percent Complete fields on the associated certification tasks.
Task Description	Information about the certification instance. This field automatically displays the text from the Task description field of the associated certification schedule.
Instructions	Field for providing instructions to the user or group performing the certification. This field is automatically filled in with information from the Instructions field on the associated certification schedule.

Users with the certification_admin role can cancel a certification instance.

Before you begin

Role required: admin

About this task

The instance must have a State of Work in Progress. Canceling a certification instance:

- Changes the certification instance State to Cancelled.
- Changes all associated Work in Progress certification tasks to Cancelled.

To cancel a certification instance:

Procedure

1. Navigate to **All > Data Certification > Schedules > Instances**.

2. Click a certification instance Number.
3. Click **Cancel**.

Use the certification task state to track the progress of a task.

Before you begin

Role required: admin

About this task

The available task states are Work in Progress, Closed Complete, Closed Incomplete, and Cancelled.

When the state of a certification task changes, the certification instance state also changes in the following cases:

- If any certification task is in Work in Progress state, the certification instance is placed in Work in Progress state.
- If all certification tasks are in Cancelled state, the certification instance is placed in Cancelled state.
- If all certification tasks are in Cancelled or Closed Complete state, the instance is placed in a Closed Complete state. For example, if three certification tasks are Cancelled, and one task is Closed Complete, the instance state is changed to Closed Complete.
- When one certification task is Closed Incomplete and the remainder of the tasks are Cancelled or Closed Complete, the instance is placed in Closed Incomplete.

To view the state of certification tasks:

Procedure

1. Navigate to **All > Data Certification > Tasks** and select **My Tasks** or **All Tasks**.

2. View the State column for each task.

Users with the certification_admin role can escalate a task in the Work in Progress state. To escalate a task, the task owner identified in the Assigned to field on the task record must have an associated manager.

Before you begin

Role required: admin

About this task

Personalize the User form to see the Manager field.

Escalating a task:

- Sends an email message to the task owner and the manager of the task owner stating that the task has been escalated.
- Sets the manager as the new task owner.

The event that triggers the escalation is named cert_task.escalate and the email notification is named Escalation Notification. To edit the text of the email message that is sent, edit the Escalation Notification email notification directly.

For more information, see [Email notifications](#).

To escalate a certification task from the Certification Task form:

Procedure

1. Navigate to **All > Data Certification > Tasks > All Tasks**.
2. Select a certification task Number.
3. Select **Escalate**.
If the Escalate button is not available, the user in the Assigned to field does not have an associated Manager.

Escalate a certification task to notify the manager of the current task owner.

Before you begin

Role required: admin

To escalate a task, the task owner identified in the Assigned to field on the task record must have an associated manager.

Procedure

1. Navigate to **All > Data Certification > Tasks > All Tasks**.
2. Select the check box to the left of a certification task Number, you can select multiple check boxes.
3. From the Actions on Selected Rows menu below the list, select **Escalate**.
If the Escalate button is not available, the user in the Assigned to field does not have an associated Manager. Select multiple tasks from the list. The menu option shows how many tasks are not eligible for escalation, such as Escalate (4 of 6).

Reassign a certification task

If you have the certification_admin role, you can reassign any certification task in the Work in Progress state. Tasks in Closed Complete, Closed Incomplete, or Cancelled state cannot be reassigned. When a task is reassigned, the current task owner and the new task owner are sent a message.

Before you begin

Role required: certification_admin

About this task

The event associated with the reassignment is named cert_task.reassign and the email notification is named Certification Task Reassignment. To edit the text of the email message that is sent, edit the Certification Task Reassignment email notification directly.

For more information, see [Email notifications](#).

To reassign a certification task:

Procedure

1. Navigate to **All > Data Certification > Tasks > All Tasks**.
2. Click a certification task Number.
3. Enter a new name in the Assigned to field.

Send certification task reminders

The Certification Task Escalations workflow sends automatic email reminders.

The Certification Task Escalations workflow sends automatic email reminders to the:

- Certification task owner.
- Assignment group, if the assignment group was specified on the Certification Task form.
- Manager of the certification task owner, if necessary and if a manager was specified on the User form.

The reminders are based on the Complete by field on the certification task record. If the Complete by date is changed, the reminder schedule automatically adjusts to reflect the new date.

Certification task reminders

Time elapsed to end date	Email reminder is sent to	Escalate field on task record reads
50%	task owner and assignment group (if specified)	Moderate
75%	task owner, assignment group, and manager of the task owner	High

Time elapsed to end date	Email reminder is sent to	Escalate field on task record reads
95%	task owner, assignment group, and manager of the task owner	High
100%	task owner, assignment group, and manager of the task owner	High

To set reminders for different or more intervals, edit the workflow Certification Task Escalations. In addition to the email reminders sent automatically, users with the certification_admin role can send email reminders manually at any time.

How to manually send email reminders from the Certification Task form.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Data Certification > Tasks > All Tasks**.
2. Click a certification task Number.
3. Right-click the header bar and select **Resend email notifications**.

How to manually send email reminders from the Certification Task list.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Data Certification > Tasks > All Tasks**.

2. Select the check box to the left of a certification task Number, you can select multiple check boxes.
3. From the Actions on Selected Rows menu below the list, select **Resend email notifications**.
4. Select multiple tasks from the list.
The menu option shows how many notifications are outstanding and how many were sent, such as Resend email notifications (15 of 18).

Mark a certification task as closed incomplete

Mark a task as closed incomplete if, for example, only some of the elements can be certified.

Before you begin

Role required: admin

About this task

The following users can mark a task as closed incomplete:

- Users with the certification_admin role.
- User identified in the Assigned to field on the certification task record.

To mark a task as closed incomplete:

Procedure

1. Navigate to **All > Data Certification > Tasks** and select **All Tasks**, or **My Tasks**.
2. Click a certification task Number.
3. In Work Notes, enter information about why the task could not be completed.
4. Click **Close Incomplete**.
If at least one task on a certification instance is marked Closed Incomplete, the Completed date and Percent complete fields on the certification instance record are not updated. A user with the certification_admin role can:

- Complete the incomplete task or tasks.

- Cancel the incomplete task or tasks.

When all tasks on the certification instance are Closed Complete or Cancelled:

- The system sets the Completed date field on the certification instance record to the current date and time.
- The Percent complete field on the certification instance record is set to 100 percent.

Certification tasks cancellation

Users with the certification_admin role can cancel a certification task in the Work in Progress or Closed Incomplete state.

When a certification task is cancelled, a notification email is sent to the task owner or assignment group assigned to the task. The task owner or assignment group manager is not notified.

Cancel a particular data certification tasks in the Work in Progress state.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Data Certification > Tasks > All Tasks**.
2. Find a task with a State of Work in Progress.
3. Click the task Number.
4. Click **Cancel**.

Cancel data certification tasks in the Work in Progress state.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Data Certification > Schedules > Instances**.
2. Find an instance with a State of Work in Progress.
3. Click the instance Number.
4. Click **Cancel**.

All tasks in the instance with a state of Work in Progress are cancelled.
The task owner or assignment group is notified.

The email template used for the notification is named `certification.task.cancelled`. The email templates can be edited to change the email message text, for example.

Domain separation and Data Certification

Domain separation is supported in Data Certification processing. Domain separation enables you to separate data, processes, and administrative tasks into logical groupings called domains. You can control several aspects of this separation, including which users can see and access data.

Support level: Basic

- Business logic: Ensure data goes into the proper domain for the application's service provider use cases.
- In the application, the user interface, cache keys, reporting, rollups, aggregations, and so on, all consider domain at production run time.
- The owner of the instance needs to be able to set up the application to function normally across multiple tenants.

Use case: When a service provider (SP) uses chat to respond to a tenant-customer's message, the client must be able to see my response.

How domain separation works in Data Certification

- Data Certification has only basic domain separation. As long as the Certification Instances (CIs) or records that must be certified are correctly domain-separated and the users who must certify the CIs or

records are in a domain that can view the data, Data Certification works as expected.

- Recommendation: The instance owner must be responsible for assigning Certification Tasks and Certification Instances to the correct domain. Changing the domain for these records does not change functionality, but limits the view of the records.

How to set up domain separation for Data Certification

After enabling the Domain Separation plugin, there are no additional steps required to set up domain separation for Data Certification.

- instance owners determine which CIs or records that need to be certified can be domain-separated.
- Customers can configure a domain-separated environment by assigning tasks to a domain, but if the data is already domain-separated, then only users with the right domain permissions can view the data in a certification task.

How tenant domains manage their own application data

It's not necessary to set the domain on the certification tables but it can be done if the instance owner should want that. As long as the CI's or records that must be certified are domain-separated, users with the correct domain permissions can view them.

Domain-separated tables

- cert_instance – Changing the domain on this table does not change any functionality, nor does it change the domains of the tasks created from the table.
- cert_task – Changing the domain on this table changes the domain viewing permissions of the task.
- cert_element – It is not recommended to change the domain on these records. As long as the CIs or records to be certified are already domain-separated, cert_element records will reflect that.
- cert_filter – Changing the domain on this table changes the domain viewing and filtering of CIs or records.

Use cases

Instance owners who have multiple clients that certify the infrastructure they own can assign domains to those CIs and the Certification Tasks to restrict the view from one client to another.

Related topics

- [Domain separation for service providers](#)

Components related to CMDB Data Manager

The following scheduled jobs, tables, roles, and properties, are related to CMDB Data Manager. Some of those, are installed with CMDB Workspace.

System properties

As a Data Manager Administrator (data_manager_admin) you can view CMDB Data Manager properties in [CMDB Workspace](#).

1. Navigate to **Workspaces > CMDB Workspace** and then select **Management** in the CMDB Workspace menu bar.
2. Select the **Data Manager** link in Management tools, in the Manage section.
3. Select **Settings** in the left navigation bar.

Properties/Settings/Labels in the Data Manager settings page

Property/Setting/Label	Description
glide.cmdb.data.manager.delete.batch.size	Size of each batch of CIs that is deleted or archived (affects performance optimization). • Type: Integer • Default value: 1000

Property/Setting/Label	Description
Retirement definitions	List of retirement definitions used in CMDB Data Manager life-cycle policies. For more information about the concept and usage of retirement definitions, see: • Retirement definitions • Manage retirement definitions for CMDB Data Manager
Scheduled job users	User group from which to select the user that scheduled jobs run as, when creating certification and attestation policies. The default value for the Scheduled job default user field is DataManager Job Runner which is a member of the default User group Data Manager Scheduled Job Users. The setting of this property determines the list of users available in the Run policy scheduled job as field on the Schedule form of a new policy.
sn_cmdb_ws.cmdb.dm.policy_types.due_date_notification Notifications	Notifications about certification and attestation incomplete tasks. When either is checked, the system notifies task owners as the due date for incomplete tasks is approaching. Notifications are sent when 50%, 70%, and 90% of the time to due date has passed. On the task due date, a past due notification is sent to the task

Property/Setting/Label	Description
	owner and to the task owner's manager. Upon closing a task, sending notifications stops. For information about how the Days to complete setting is used to calculate when notifications are sent, see Create a CMDB Data Manager policy.
sn_cmdb_ws.glide.cmdb.data.manager.aging.policy.task.timeframe Aging policy task timeframe	Number of days before a Data Manager task is marked as aged. - Type: Integer - Default value: 30
glide.cmdb.data.manager.subflow.timeout Threshold for subflow run time	Threshold (in milliseconds) for subflow running time. A subflow that passes this threshold while running, is cancelled. - Type: Integer - Default value: 600,000 (10 minutes)
glide.cmdb.data_manager.default_archive_time Number of days CIs are archived before they get deleted permanently	Number of days that it is still possible to restore archived CIs from archive tables. After the specified number of days pass, archived CIs are permanently deleted from the archive tables. Type: Integer

Property/Setting/Label	Description
	Default value: 120
cmdb.data.manager.stale.task.life.in.days Number of days after which task is considered stale	Number of days after which a task is considered stale and is set to Closed Cancelled by the CMDB Data Manager - Stale Task Cleaner daily scheduled job. Details: - Type: Integer - Default value: 90
sn_cmdb_ws.glide.cmdb.data_manager.rejected.task.notifications.group Group to receive notifications for a rejected task	System identifier of a user group to notify about tasks that are being rejected. By default, this property has no value, in which case, all data manager administrators are notified about rejected tasks. For information about how this property is used when reviewing and rejecting life-cycle tasks, see Review CMDB Data Manager tasks.

Scheduled jobs

Scheduled job	Description
CMDB Data Manager Archive/Delete Policy Processor	Processes all published policies of type Archive and Delete:

Scheduled job	Description
	- Applies policies only to CIs that are already retired. - Processes all Archive policies first, and if no errors encountered, continues to process any Delete policies.
CMDB Data Manager Retire Policy Processor	Processes all published policies of type Retire: Applies policies only to CIs that are not retired.
CMDB Data Manager - Stale Task Cleaner	Cleans up stale CMDB Data Manager tasks by setting the task to Closed Cancelled. The <code>cmdb.data.manager.stale.task.life.in.days</code> system property determines the number of days after which a task is considered stale (90 by default). A task becomes stale if: - The task was created at least 90 days (by default) ago and it is still open. - The approval requests are older than 90 days (by default) and the task is not yet approved.
CMDB Data Manager Delete Related Entry Policy Processor	Processes the Delete CMDB Related Entry policy by deleting the specified related tables from the Related Entry [cmdb_related_entry] table.

Tables

Table	Description
CMDB Data Management Policy [cmdb_data_management_policy]	Details about CMDB Data Manager policies.
CMDB Data Management Policy Executions [cmdb_data_management_policy_execution]	Execution records that correspond to each policy run. As policies run according to their recurring schedule, for each run the system generates a record with a unique policy execution number for that run. A policy execution number has a prefix of 'PLCEXEC'. Execution records associated with certification policy runs, are used for populating the Data Certification Dashboard.
CMDB Data Management Policy Runtime Attributes [cmdb_data_management_policy_runtime_attributes]	Current policy metadata including status and summary.
CMDB Data Management Task Control [cmdb_data_management_task]	Open policy tasks generated by published policies.
CMDB Data Management Tasks to CIs	Associations of tasks to CIs.

Table	Description
[cmdb_data_management_task_to_ci]	
CMDB Data Manager Task to Documents [cmdb_data_management_task_to_document]	Associations of task and to related items.
CMDB Data Management Attestation Tasks to CIs [sn_cmdb_ws_attestation_task_to_ci]	Associations of task and attestation CIs.
CMDB Data Management Tasks to CIs [cmdb_ci_state_management_task_to_ci]	Associations of task and CIs for 'Dependent CI Deletion' policy type.
Excluded CIs [cmdb_policy_ci_exclusion_list]	Tracks the CIs that are set to be excluded during policy evaluation.
CMDB Policy Type [cmdb_policy_type]	Policy types supported by the CMDB Data Manager.
CMDB Policy Type Categories [cmdb_policy_type_categories]	Associations of policy types to Workflow Studio Categories.

Table	Description
CMDB Retirement Custom Definitions [cmdb_retirement_custom_definitions]	Custom retirement definitions used in life-cycle policies (Retire/Archive/Delete).
CMDB Policy Scheduled Jobs [cmdb_policy_scheduled_job]	Policy scheduled Job details.
Dependent CI Relations Evaluation Config [cmdb_dependent_ci_extra_rels_config]	Information about CIs' extraneous relationships which is used for the delete and archive operations. Extraneous relationships are those relationships that are not in the CI's chain of dependency. For more information, see Dependent CIs management .
CMDB Dependent CI Class Exclusions [cmdb_dependent_ci_class_exclusion]	Excluded classes for dependent CI, which is used with the Archive/Retire/Delete policy types to determine which CIs are orphan dependent CIs. For more information, see Dependent CIs management .
CMDB Dependent CI Ledger [cmdb_dependent_ci_ledger]	Staging table for orphaned dependent CI related to the deletion or archival of the main CI that they depend on. Also used in a cascade-retire dependent CIs operation.

Table	Description
	For more information, see Dependent CIs management .
CMDB CI End Of Life Ledger [cmdb_ci_end_of_life_ledger]	CIs that are being set as retired that belong to a class that has any downstream dependencies. Used in the Dependent CI Retire policy.

Roles

Role title [name]	Description	Contains roles
CMDB Data Manager administrator [data_manager_admin]	Can access all features in the CMDB Data Manager, including: • Full access to assigned tasks. • Full access to policies. • Ability to associate subflow categories to policy type. Can create, edit, and delete policies, calculate previews, approve tasks, and manage exclusion lists.	• task_editor • data_manager_user • cmdb_query_builder_read

Role title [name]	Description	Contains roles
CMDB Data Manager user [data_manager_user]	Can view CMDB Data Manager policies in read-only mode and calculate previews. Can perform the following tasks: • View assigned tasks. • Update, approve, or reject an assigned task. • Add a CI to an exclusion list from their task.	cmdb_read
DataManager Job Runner [dataManager.jobRunner]	Default user for the Scheduled job default user CMDB Data Manager property. The Run policy scheduled job as field on the Schedule form in a policy, is set to this user if no other option is chosen.	